

Captured weight and boundary leakage bound the error of sample-based spectral functions

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Rayleigh–Ritz on a subspace from sample-based quantum diagonalization yields the spectral function of a probe state; we ask what bounds its error. The probe’s captured Born weight w does not do so alone: an analytic counterexample forces any bound indexed on it to its trivial value. On Hubbard rings at broadening $\eta = 0.18t$, the probe’s whole support ($w = 1$) still gives a relative error of 0.35–0.43: Rayleigh–Ritz displaces the poles a subspace keeps. The error is bounded instead by the missed weight and the Hamiltonian coupling across the subspace boundary. Given the exact ground state, we prove for any orthogonal projector that the relative L_1 error lies between $1 - w$ and $\min\{1 + w, (1 + \sqrt{w})(\sqrt{1 - w} + \hat{\Lambda}(\eta)/\eta)\}$, where $\hat{\Lambda}$, that coupling at resolution η , is computed from the Ritz pairs and the retained probe. We test it on L -site rings with classical Born draws of an exactly time-evolved probe, or their infinite-shot ranking; our device runs are execution records only. On the ranked subspaces the bound is vacuous at every operating fraction of the resource scan, at all five sizes up to a sector of 10 306 296 determinants: its leakage branch exceeds the trivial bound $1 + w$ by factors of 1.66 to 8.61. Only well above those fractions is it informative, and there it is calibrated: at $L = 6$ –12 it first beats the trivial bound at a true error of $(1.2\text{--}7.3) \times 10^{-3}$, though an unproven fit in $1 - w$ is tighter on 11 of 14 splits. For the Born-ranked $A(k, \omega)$ at 85% of its sector, the bound is 350–900 times the true error. In the determinant basis, zero leakage at full weight needs every symmetry-allowed determinant: the probe’s measured Krylov support at $L = 6$ –14. Plane-wave orbitals reduce a momentum probe’s support to one block, $1/L$ of an exponentially growing sector. At full weight a second-order certificate follows; whether one exists below full weight, where every operating fraction lies, is open.

I. THE OPERATIONAL QUESTION

A sample-based quantum calculation returns a list of bitstrings. Diagonalizing the Hamiltonian in the span of the configurations they name yields an energy and, as this work and several others show, a frequency-resolved response as well. What it does not yield is a statement of how wrong that response is. This paper addresses the question a user faces once the shots are spent: *given the subspace those shots populated, how wrong is the spectral function built on it?*

The natural diagnostic is the captured Born weight, the fraction of the probe’s squared norm that lies in the subspace; the probe is the ground state acted on by the operator whose response is sought. It does not answer the question. Rayleigh–Ritz does not delete the poles a subspace misses; it displaces the poles it keeps. A subspace that holds the probe’s entire Born support, where the captured weight is exactly one, therefore still carries an error of order one. On half-filled Hubbard rings of L sites, at Lorentzian broadening $\eta = 0.18t$, the relative L_1 error is 0.43 at $L = 6$ and 0.35 at $L = 8$ (0.28 on the open chain of Sec. IV, a different lattice). No choice of constant rescues the captured weight: an analytic counterexample forces the constant of any bound indexed on it to its trivial value (Sec. IIID).

What enters a bound, together with the missed weight, is the Hamiltonian coupling across the subspace boundary. Theorem 1 makes this precise for Rayleigh–Ritz on any orthogonal projector: the L_1 error of the broadened spectral function lies between the missed Born weight and the smaller of the trivial bound and a leakage branch, built from that weight and the coupling that leaks across the boundary at resolution η . The leakage term is a posteriori: it has a closed form in the Ritz pairs the reconstruction produces and in the retained probe. Both terms take the probe and the energy reference from the exact ground state, so the bound is a certificate for benchmarking a sample-based pipeline where exact diagonalization is available. The identity behind it also shows that the error surviving at full weight is generated by the boundary coupling alone (Sec. IIID), whatever value the bound itself takes there.

What the subspace structure does guarantee is moment exactness, the classical moment-matching property of Krylov projections (Sec. IV). A subspace containing the order- K Krylov space reproduces the spectral moments through order $2K + 1$ and no further; we state that reach exactly and measure it on coordinate subspaces (Theorem 2). A non-circular count then gives the number of shots needed to capture every high-probability configuration (Proposition 1).

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We then measure where the bound is informative. The operating fractions of the resource scan are its acceptance fractions: the fractions of the symmetry sector that the infinite-shot ranking of the scan of Sec. VII needs to meet this paper’s 5% accuracy criterion, relative $L_1 < 0.05$ at $\eta = 0.15t$. The leakage branch of the bound is worse than the trivial bound at every operating fraction of the resource scan, at five sizes up to a sector of dimension 10 306 296, by factors of 1.66 to 8.61. At $L = 12$ that verdict is converged in Krylov depth. The bound becomes informative only well above those fractions, and to certify the 5% criterion at $\eta = 0.18t$ it needs at least 0.98 of the sector at $L = 8$. An independent implementation recovers the acceptance fractions of the scan to under 2% and confirms the 5% criterion on the published subspaces: the reconstructions are reproduced, and what they lack is a certificate.

Where the bound is informative, it is calibrated. Over sixteen (L, η) cells at $L = 6$ –12, it first crosses below the trivial bound at a true relative error between 1.2×10^{-3} and 7.3×10^{-3} : a spread of 5.9, over true errors that span five decades. Once it is below that bound, the true error is already at that level. A two-constant rule in the missed weight, fitted on some sizes and tested on the others, localizes this crossing more tightly on eleven of fourteen splits. That rule is a regularity of these systems with no guarantee outside its fitted range; the certificate is a proven inequality, given the exact probe. The Born-ranked (infinite-shot) $A(k, \omega)$ of Sec. V, at 85% of its sector, is non-vacuous in all sixteen momentum channels. Its relative bound there, 0.77–1.75 against the trivial 2, is 350–900 times its true error.

The leakage depends on the basis in which the device measures. In the determinant basis, the Krylov space of a probe created at one site, the seed site, has coordinate support on every determinant that the reflection about the seed site allows, measured at $L = 6$ to 14. At full captured weight, therefore, only a subspace holding all of them has zero leakage. If the device samples in plane-wave orbitals instead, a momentum-resolved probe has its Krylov support in one momentum block, $1/L$ of the sector. The rotation to that basis costs about $L(L - 1)$ further two-qubit gates, whose effect on the sampled distribution is unmeasured, and the block, like the sector, grows exponentially with L (Sec. VE).

No orbital-rotation invariant can bound the boundary leakage usefully on every state: on fibers of a pairing model, families of states on which the invariants are held fixed, every such bound is vacuous. None can therefore replace the certificate as an a priori predictor computed from the state before any circuit runs (Sec. VI). The natural candidates are the efficiently measurable monotones of fermionic non-Gaussianity. On number-conserving states the one-body member collapses to the single invariant $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$, with γ the one-particle reduced density matrix [Eq. (17)]. An invariant of the occupation spectrum cannot see a boundary. Along a geminal orbit every antiflatness $\mathcal{F}_k$, like every other orbital-rotation invariant, is fixed, while the orbit contains a point of classical cost $O(K)$, for K geminals, in the fixed computational basis (Theorem 3). That basis is the pairing basis, in which the point has 2^K determinants but bond dimension 2. The orbit also contains points at which Born sampling is plausibly hard (Sec. S5), a hardness we do not establish. On those fibers the captured weight is frozen too, yet the subspace-aware certificate still separates a non-vacuous value from the trivial bound (Proposition 2).

Within each of the six (L, N) strata of the Hubbard interaction sweep, the rank correlation of $\mathcal{F}_1$ with the determinant support is exactly -1 . Its exact permutation $p = 3.35 \times 10^{-13}$ assumes independent orderings, a null that six strata sharing one mechanism do not satisfy. The correlation is a deterministic monotone relation whose sign reverses between families, as it must when an orbital invariant is compared with a basis-dependent count (Sec. VI).

Sections VII and VIII then price the method in support, resolution and shots, and check the full-sector references against the opposite finite-size scaling of the charge and spin gaps of the Hubbard chain.

Four bodies of work bound what is new here. (i) Patel, Rangi and Tam [1] construct Green’s functions from sampled Krylov subspaces for dynamical mean-field theory [2], independently and concurrently, and cite this work as “a closely related sample-based approach to dynamical spectral functions, constructed directly from bitstring-sampled subspaces”. (ii) Dynamical DMRG computed all three momentum-resolved channels reported here for the half-filled extended Hubbard chain at $L = 60$ –120 in 2007 [3]. Correction-vector DMRG treats the Hubbard chain at $L = 48$, the Heisenberg chain at $L = 64$ (the $k = \pi$ cut to $L = 128$) and the t – J ladder at 48×2 sites [4], and the classical toolbox for such spectra is broader still [5–10]. (iii) Frequency-resolved responses of lattice models, all but one of them correlated, have been measured on devices far larger than the 12-qubit spectral run of this work. They include a retarded spin Green’s function of the Fermi–Hubbard chain at up to 30 qubits [11], an angle-resolved-photoemission-like $A(k, \omega)$ of a free-fermion 27-site chain at 54 qubits [12], $S(q, \omega)$ on 50 qubits against neutron scattering in KCuF_3 [13], and quench spectroscopy of $L = 101$ spins [14]. At a scale comparable to ours, $S(q, \omega)$ of a 20-site Heisenberg chain was computed on a trapped-ion device and compared with neutron data for copper sulfate [15]. The two device runs of this work are execution records only: the spectral one retained its entire sector, so its agreement with exact diagonalization is exact by coverage rather than a test of the method (Sec. IX A). (iv) Computable Gaussian-invariant quantifiers of fermionic non-Gaussianity include the antiflatness of Ref. [16] used here. Tarabunga *et al.* [17] separate covariance-matrix occupation entropies from full-state natural-orbital participation entropies. Two members of the first family are strong monotones that lower-bound the non-Gaussian gate count; the second family upper-bounds the classical simulation cost in an orthonormal Gaussian basis. The quadratic member, which coincides with the $k = 1$ antiflatness, is one of the two strong monotones. Leone and Bittel [18] independently prove it an efficiently measurable strong pure-state Gaussian monotone. Tarabunga *et al.* and Leone and Bittel posted their works both in July 2026; the priority for those

measures is theirs. What this work adds is the contrast between these measures and the basis-dependent determinant support, a constructive instance of that contrast, and its extension from the determinant support to the leakage. Every system studied here is classically tractable: Hubbard rings at $L \leq 14$ and nineteen active spaces with a largest minimal bond dimension of 42. No quantum advantage is claimed. The one quantitative statement this work makes about its own cost—that the basis-dependent bond dimension tracks the determinant support—argues, wherever that bond dimension is small, for a tensor network rather than for a sampler (Sec. IX).

The paper is organized as follows. Section II fixes the sampling primitive and separates shot-budgeted reconstructions from full-sector references. Section III proves the bound and shows that no bound on the captured weight alone can replace it; Sec. IV states what the subspace structure certifies; Sec. V measures where the certificate is informative; Sec. VI addresses a priori predictors; Secs. VII and VIII price the method and validate the references; and Sec. IX reports the device runs, the molecular suite and the relation to classical methods. The systems are the half-filled one-dimensional Hubbard model at $U/t = 8$, rings and open chains at $L = 4$ to 14, and nineteen molecular active spaces. Three quantities share the symbol $\mathcal{S}$: the static ground-state support of Sec. VI, the dynamical support of Sec. VII, and the set of configurations the sampler returned, the $\mathcal{S}$ of Theorem 1. The proof of Theorem 1 is in Appendix A; the other proofs, full tables and extended discussions are in the Supplemental Material [19], cited as Sec. Sn.

II. ONE SAMPLING PRIMITIVE, FOUR CHANNELS: WHICH ARE RECONSTRUCTED AND WHICH ARE REFERENCES

The quantum processor is asked for one thing: computational-basis bitstrings of a time-evolved response seed. The subspace those bitstrings name, its diagonalization and every spectral function built on it are computed classically, and Table I states, channel by channel, which objects here were reconstructed from measurements and which are full-sector references.

A. Construction

For a Hamiltonian $\hat{H}$ with N -electron ground state $|0\rangle$ of energy E_0 , the single-particle spectral function has an electron-addition and an electron-removal branch,

$$A_p(\omega) = \sum_n |\langle n^{N+1} | \hat{c}_p^\dagger | 0 \rangle|^2 \delta(\omega - (E_n^{N+1} - E_0)) + \sum_m |\langle m^{N-1} | \hat{c}_p | 0 \rangle|^2 \delta(\omega - (E_0 - E_m^{N-1})), \quad (1)$$

energies measured from E_0 ; the chemical potential $\mu = \frac{1}{2}(E_0^{N+1} - E_0^{N-1})$ places the Fermi level mid-gap for the particle-hole-symmetric Hubbard model at half filling. The subspace is built in three steps. (1) Prepare the response seeds $|\phi^\pm\rangle = \hat{c}_p^\dagger|0\rangle, \hat{c}_p|0\rangle$ in the $(N\pm 1)$ sectors (every simulation here takes $|0\rangle$, the seeds and $e^{-i\hat{H}t_k}$ exact; how the non-unitary seed is prepared on a device is not specified by this work). (2) Time-evolve each, $|\phi^\pm(t_k)\rangle = e^{-i\hat{H}t_k}|\phi^\pm\rangle$, and measure computational-basis bitstrings at each of the $K+1$ snapshots; the union of the *distinct* bitstrings returned is the determinant subspace $\mathcal{S}^\pm$. (3) Form $\hat{H}_{\mathcal{S}^\pm} = P_{\mathcal{S}^\pm} \hat{H} P_{\mathcal{S}^\pm}$, diagonalize it as $\sum_m \tilde{E}_m^\pm |\tilde{m}^\pm\rangle \langle \tilde{m}^\pm|$, and evaluate the Lehmann sum inside $\mathcal{S}^\pm$. The retarded Green's function is then assembled entirely on the classical processor,

$$G_{pq}(\omega) = \sum_m \frac{\langle 0 | \hat{c}_p | \tilde{m}^+ \rangle \langle \tilde{m}^+ | \hat{c}_q^\dagger | 0 \rangle}{\omega - (\tilde{E}_m^+ - E_0) + i\eta} + \sum_m \frac{\langle 0 | \hat{c}_q^\dagger | \tilde{m}^- \rangle \langle \tilde{m}^- | \hat{c}_p | 0 \rangle}{\omega - (E_0 - \tilde{E}_m^-) + i\eta}, \quad (2)$$

whose imaginary part is the Lorentzian-broadened $A_{pq}(\omega)$ at resolution η . The seed selects the channel: charged seeds $\hat{c}_q^\dagger|0\rangle$ in the $(N\pm 1)$ sectors give $A(\omega)$ and, by Fourier transform of G_{pq} , $A(k, \omega)$; number-conserving density and spin seeds $\hat{n}_q|0\rangle, \hat{S}_q^z|0\rangle$ in the neutral N sector give $S(q, \omega)$ and $S^{zz}(q, \omega)$. Figure 1 schematizes the primitive—what one circuit and one readout can address; Table I states separately what was actually run for each channel here.

B. Relation to quantum subspace and Krylov methods

For equally spaced snapshots the span of the $K+1$ time-evolved seeds is the order- K Krylov space of the propagator $e^{-i\hat{H}\Delta t}$ [23], a real-time counterpart of the polynomial Krylov space $\text{span}\{\phi, \hat{H}\phi, \dots, \hat{H}^K\phi\}$, which block encoding

TABLE I. **Provenance of every dynamical object in this paper: the subspace it was built on, and the measurements that built it.** “Full sector” means no subspace selection of any kind—the continued fraction runs on the whole sector at the stated recursion depth n_ℓ , so the row is a reference against which reconstructions are scored, not a reconstruction. “Observed bitstrings” means the subspace is the union of the distinct bitstrings actually returned by T draws split evenly over the $K+1$ snapshots, which is the protocol of Sec. II; one *stream* is one independent pseudorandom realization of that draw. “Born ranking” means the subspace is the top fraction of the sector under the accumulated Born weight of the evolved state, which is the $T \rightarrow \infty$ limit of the same protocol and carries no shot budget. Sector dimensions are exact; budgets are per channel and per stream.

Channel and system	Shown in	Determinant subspace	Shots T
$A(\omega)$; open Hubbard chain $L=6$, $U/t=8$, $\eta=0.15t$; dim = 300	Fig. 2	observed bitstrings, plus the largest-amplitude determinant of the probe, inserted and not drawn; $ S = 245$ of 300 in panel (a), 242.5 on average in panel (b)	5.2×10^4 per stream (4×10^3 at each of 13 snapshots, $K=12$), 8 streams
$A(\omega)$, orbital projected; 19 molecules at dissociation; dim = 2–300	Table S12	observed bitstrings, plus the largest-amplitude determinant of the probe, inserted and not drawn; the fraction $ S /D$ is the last column of that table	2.16×10^5 per molecule (8×10^3 at each of 27 snapshots, $K=26$), one stream
$A(\omega)$; N_2 along dissociation, CAS(6,6)	Fig. S6(a)	full CAS sector	none
$A(\omega)$; hardware, <code>ibm_fez</code> , $L=6$, $U/t=4$, $\eta=0.15t$; dim = 300	Fig. 9(a)	post-selected from device bitstrings read in reversed bit order, so every retained shot is a device error (Sec. IX A); $ S = 300$ of 300, the entire sector, so the reconstruction equals the exact $A(\omega)$ by coverage	3.5×10^5 : 5×10^4 on each of 7 circuits, $k=0$ (the $t=0$ reference) to $k=6$
$A(k, \omega)$; $L=12$, $\eta=0.18t$; dim = 731 808	Fig. 4	full sector, $n_\ell = 150$ per momentum	none
$A(k, \omega)$; $L=8$, $\eta=0.18t$; dim = 3920	Fig. 3(a,b)	Born ranking, 3332 of 3920 per channel, over 17 snapshots ($K=16$)	none: (a)–(b) are the $T \rightarrow \infty$ limit; at finite shots 5.2×10^6 per channel suffice and 2.6×10^6 do not (see text)
$A(\omega)$; $L=8$, $\eta=0.15t$; dim = 3920	Fig. 3(c)	observed bitstrings; sector fraction 0.835 ± 0.0025 at the largest budget	up to 2.60×10^6 over 19 snapshots ($K=18$), 8 streams
$S(q, \omega)$; $L=12$, $\eta=0.18t$; dim = 853 776	Fig. 7	full N -particle sector, $n_\ell = 200$	none
$S^{zz}(q, \omega)$; $L=12$, $\eta=0.10t$; dim = 853 776	Fig. 8	full N -particle sector, $n_\ell = 200$	none
Required sector fraction versus L ; $L=4$ –14, $\eta=0.15t$	Fig. 5(b,c)	both selections, plotted together: Born ranking (open squares) and observed bitstrings (filled circles)	measured: 2.4×10^4 , 2.6×10^5 , 3.1×10^6 , 1.9×10^7 at $L = 6, 8, 10, 12$ (Table IV, 4–6 seeds per size)

builds exactly without time evolution [24]; diagonalizing $\hat{H}$ inside a subspace containing such a space is the route of quantum subspace expansion [25], filter diagonalization [26] and real-time quantum Krylov algorithms [27], whose error theory is now characterized [28]. What distinguishes the present method is that these states are used not to *estimate matrix elements* [29–34]—the step requiring Hadamard tests, compute–uncompute overlaps, Hamiltonian-moment or quantum equation-of-motion measurements on the device—but as a *sampling distribution* selecting a determinant subspace [35–41], extended since to embedded fragments and to auxiliary-field post-processing [42, 43], and here lifted to the $(N \pm 1)$ manifold and to a frequency-resolved observable. Apart from the concurrent Green’s-function extension of Ref. [1] (Sec. I), that lineage stops at energies, static correlators and, at most, discrete levels: the closest published results are a QSCI quasiparticle band structure returning discrete poles [44], a molecular spectrum built from a neutral-sector autocorrelation with classically projected long-time dynamics [45], and, from the measured-overlap side rather than the sampled one, a dynamical structure factor of a Kitaev spin liquid obtained by quantum subspace

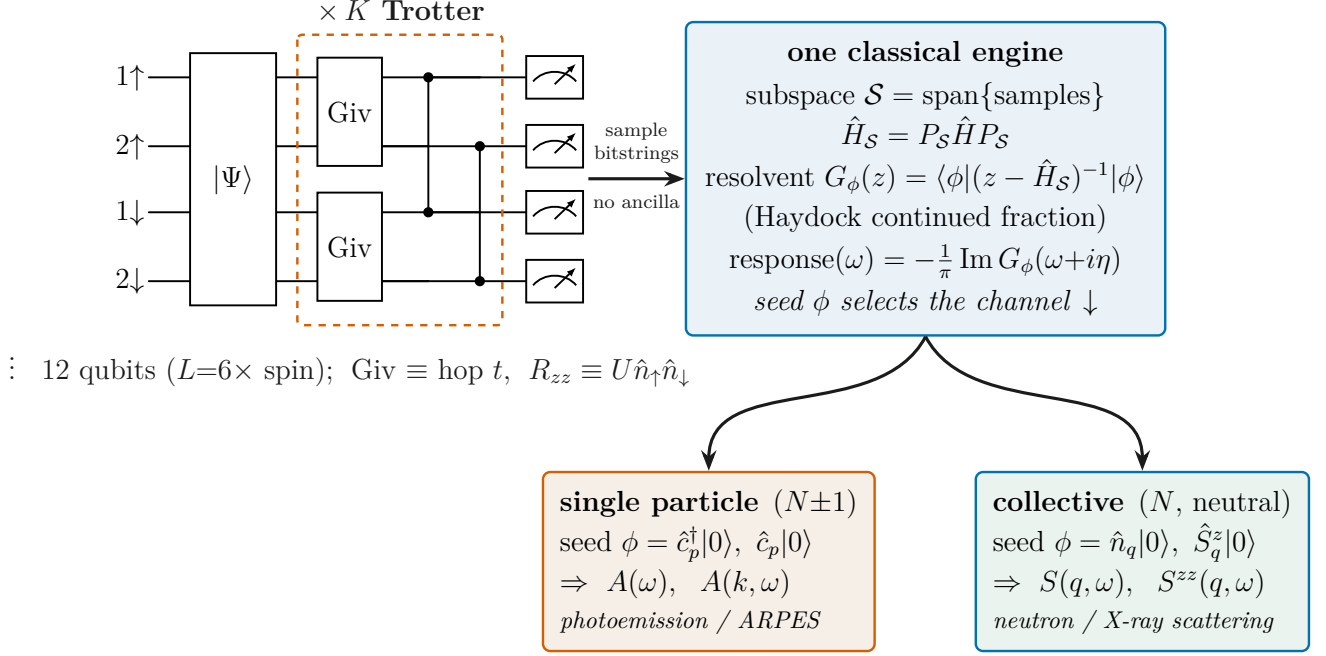


FIG. 1. One sampling primitive, four dynamical responses. An ancilla-free circuit—state preparation and a Trotter layer repeated K times (Givens rotations for the hopping t , native R_{zz} for the on-site $U \hat{n}_\uparrow \hat{n}_\downarrow$)—is measured in the computational basis; no Hadamard test or controlled unitary is run on the device. The sampled bitstrings define a sector-specific subspace $\mathcal{S}$ and feed a single classical engine—a Haydock continued-fraction resolvent [20] $G_{\phi}(z) = \langle \phi | (z - \hat{H}_{\mathcal{S}})^{-1} | \phi \rangle$, whose single-particle case ($\phi = \hat{c}_p^\dagger |0\rangle$) is the Green’s function of Eq. (2). The seed ϕ then selects the channel: particle addition/removal seeds $\hat{c}_p^\dagger |0\rangle, \hat{c}_p |0\rangle$ in the $(N \pm 1)$ sectors give the single-particle spectral functions $A(\omega)$ and $A(k, \omega)$ (photoemission/ARPES), while number-conserving density and spin seeds $\hat{n}_q |0\rangle, \hat{S}_q^z |0\rangle$ in the N sector give the dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$ (neutron [21] and resonant inelastic X-ray [22] scattering)—all from *one* sampling primitive, the seed selecting the sector and hence the subspace. Here $A(\omega)$ is reconstructed from sampled bitstrings (Fig. 2) and $A(k, \omega)$ on a Born-ranked subspace, the infinite-shot limit of the protocol (Fig. 3); the $A(k, \omega)$ of Fig. 4, $S(q, \omega)$ and $S^{zz}(q, \omega)$ are full-sector references, not reconstructed here (Table I).

expansion [46], or a hybrid quantum–classical framework for many-fermion response and structure [47]. Closest on lattice models, Nogaki *et al.* [48] apply symmetry-adapted SQD to ground- and excited-state *energies* and a *static* superconducting pair correlation of the two-leg Hubbard ladder, and Wray *et al.* [49] measure how SQD converges on a variable-length cuprate chain. The readout is that of SQD, the circuits are the Trotterized Krylov evolutions of Krylov quantum diagonalization [50], sampled as in sample-based Krylov diagonalization [37, 51], and all dynamical information is assembled classically through Eq. (2) by one continued-fraction resolvent [20, 52].

C. What the subspace reconstruction is: Rayleigh–Ritz, not restriction

Step (3) is the definition the rest of this paper turns on. With $P_{\mathcal{S}}$ the orthogonal projector onto the span of the determinants in $\mathcal{S}$ and ϕ the seed,

$$A_{\mathcal{S}}(\omega) = \frac{\eta}{\pi} \left\| (\omega + E_0 + i\eta - H_{\mathcal{S}})^{-1} P_{\mathcal{S}} \phi \right\|^2, \quad H_{\mathcal{S}} = P_{\mathcal{S}} \hat{H} P_{\mathcal{S}}, \quad (3)$$

the spectral function of the *Rayleigh–Ritz operator* $P_{\mathcal{S}} \hat{H} P_{\mathcal{S}}$, whose poles are the Ritz values $\tilde{E}_m$ of Eq. (2)—not eigenvalues of $\hat{H}$ —and not the restriction of A to a subset of its own eigenpairs. The reproduction repository computes exactly this, at the small sizes by forming and diagonalizing the submatrix $\tilde{H}[\mathcal{S}, \mathcal{S}]$ and at the large ones by a matrix-free restricted matrix–vector product inside the same continued fraction; the consequence is the mechanism of this paper: *spectral weight the subspace retains is not kept in place, it is moved in ω* . Truncation does not delete Lehmann poles, it displaces every surviving one, so the weight a subspace captures cannot by itself bound how far the reconstruction has moved—which is why no weight-only bound survives contact with Eq. (3) (Sec. IIID), and why Theorem 1 carries a

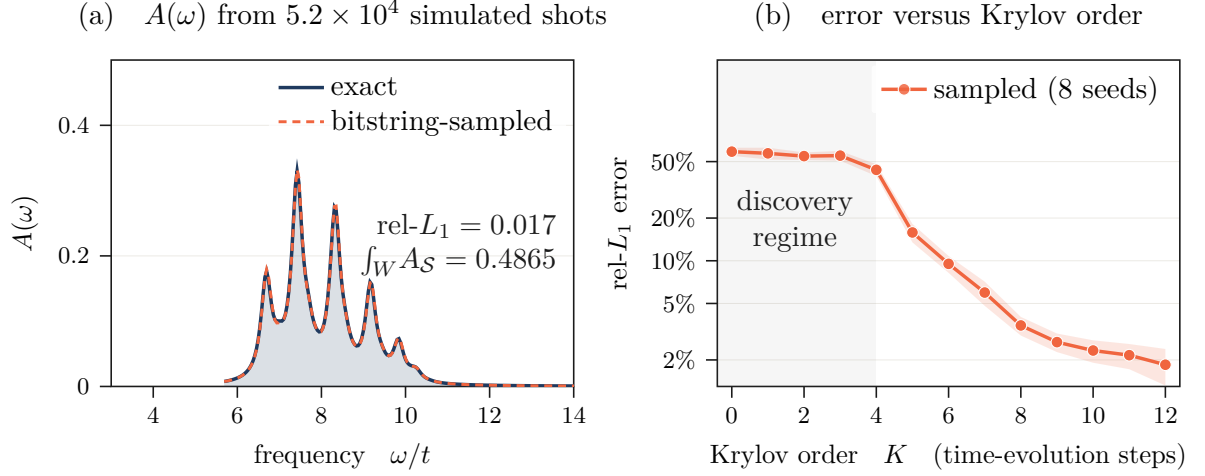


FIG. 2. **The method validated at $L = 6$, and its error against Krylov order.** (a) The single-particle addition branch $A^+(\omega)$, reconstructed from classically simulated shots (independent draws from the exact Born distribution of the evolved probe, with the probe’s largest-amplitude determinant inserted into each subspace rather than drawn), reproduces the exact one (open Hubbard chain, $L = 6$, $U/t = 8$; broadening $\eta = 0.15t$) to relative- L_1 0.017—the exact (filled) and reconstructed (dashed) curves are indistinguishable. The analytic addition-branch sum rule $\int A^+ d\omega = 1 - \langle n_p \rangle = 0.5$ is an identity of the ground state and not a check on the reconstruction; integrated over the plotted window the exact and reconstructed curves give 0.486588 and 0.486471, agreeing with each other to 2.4×10^{-4} relative and both falling 2.7% short of 0.5 because Lorentzian tails leave the window (Sec. S7). (b) Convergence of the relative- L_1 error with the Krylov order K (number of time-evolution steps sampled), seed-averaged over eight independent runs (band = standard deviation), $\eta = 0.15t$, so the budget is $T = 4000(K+1)$ and reaches 5.2×10^4 at $K = 12$. The error falls by a factor 32 across the thirteen points, from 0.59 at $K = 0$ to 0.0185 at $K = 12$, with one non-monotone step at $K = 3$ inside the discovery phase, while the mean subspace returned grows from 75 to 242 determinants of the 300-dimensional sector. No geometric rate is fitted to the window $4 \leq K \leq 9$ here (Sec. IV B): one is not implied by moment exactness, and inside that window the decay in K is confounded with the growth of $|\mathcal{S}|$.

second term indexed on the coupling $P_{\mathcal{S}} \hat{H} Q_{\mathcal{S}}$ across the boundary of $\mathcal{S}$, with $Q_{\mathcal{S}}$ the complementary projector—exactly what Eq. (3) discards. Two consequences used later, that determinants outside $\text{supp}(\phi)$ still move the Ritz spectrum and that the error does not vanish on $\text{supp}(\phi)$ itself, are stated in Sec. S1 and measured in Sec. III D.

D. Provenance: what was reconstructed from shots, channel by channel

Table I gives, for every dynamical object in this paper, the subspace it was built on and the measurements that built it. The neutral channels $S(q, \omega)$ and $S^{zz}(q, \omega)$, the $A(k, \omega)$ of Fig. 4 and the N_2 series of Fig. S6 are full-sector references computed by continued fraction, not reconstructions. The Born ranking of Fig. 3(a,b) and of the published curve of Fig. 5 is the infinite-shot limit of the declared protocol, and its finite-shot replacement is measured with its budget: at $L = 8$, 2.60×10^6 shots over 19 snapshots reach sector fraction 0.835 ± 0.0025 at relative L_1 error $(4.81 \pm 0.27) \times 10^{-3}$ (Fig. 3(c)), and the literal configuration of Fig. 3(a,b) reaches the accuracy of panel (b) at 5.2×10^6 shots per channel and not at half that budget. The full declarations, including the branch sum rules and what is guaranteed in advance, are in Sec. S1.

III. TWO BUDGETS: THE MISSED WEIGHT AND THE BOUNDARY

The L_1 error of a spectral function reconstructed on a sampled subspace lies between the Born weight the subspace missed and that weight plus a boundary-leakage term, $\Lambda_P(\eta)/\eta$, computed from the Ritz pairs the reconstruction has already produced and one matrix–vector product per Ritz vector (Theorem 1). The leakage term does not need the exact A ; it does need the exact ground state (H4). The bound is two-sided, it holds for *any* orthogonal projector and therefore for any set of configurations a sampler might return, and its upper branch has two terms: the first is fixed by the captured weight and vanishes once the probe’s support is held; the second is not controlled by the weight, although it falls as the subspace grows. Section V measures where the resulting certificate is informative, Sec. VI shows that no invariant of the state can replace it, and Sec. VII prices both terms in shots.

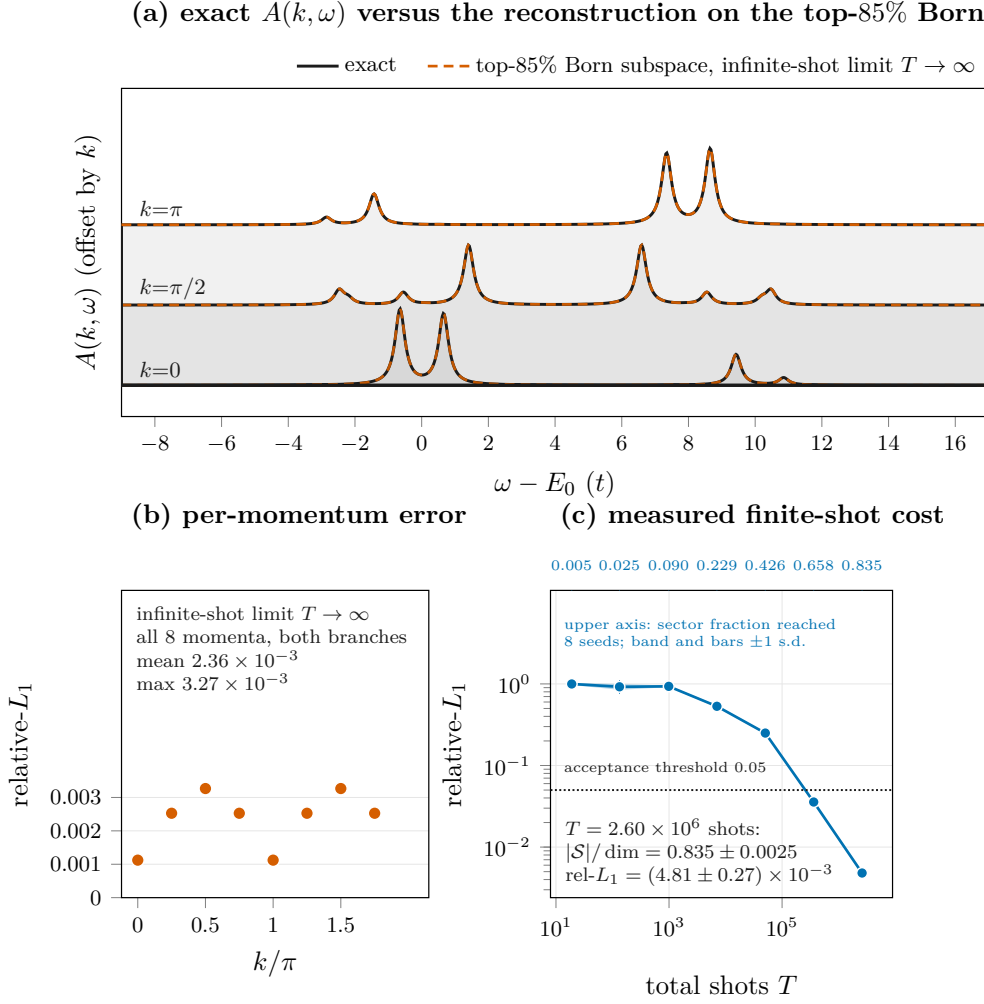


FIG. 3. **The momentum-resolved spectral function on a Born-ranked subspace, and what that subspace costs in shots.** (a) Exact $A(k, \omega)$ (solid, shaded) and the reconstruction obtained by diagonalizing the top-85% of the $(N \pm 1)$ sector under the time-averaged Born weight of the evolved state (dashed), at three momenta ($k = 0, \pi/2, \pi$; Hubbard ring, $L = 8$, $U/t = 8$, $\eta = 0.18t$, $K = 16$ time slices, $\delta t = 0.5$; curves offset vertically by k). Frequencies are measured from E_0 with no chemical-potential shift, so the removal branch lies below $\omega - E_0 = 1.41t$ and the addition branch above $6.59t$; the plotted and scored window, $\omega - E_0 \in [-9t, 17t]$, is symmetric about $\mu = U/2$ and holds both branches. The subspace is 3332 of the 3920 determinants of each channel. The ranking is taken on the *exact* Born distribution, so panels (a) and (b) are the **infinite-shot limit** $T \rightarrow \infty$ of the protocol, not a finite-shot experiment; no shot budget is attached to them. (b) Relative- L_1 error of that reconstruction at all 8 physical momenta, addition and removal branch summed over that window: mean 2.36×10^{-3} , maximum 3.27×10^{-3} . (c) What a finite shot budget buys, measured. Same ring and the same sector dimension ($L = 8$, $\dim(N+1) = 3920$) but a separate run: local seed $c_{0\uparrow}^\dagger |\Psi_0\rangle$, $\eta = 0.15t$, $K = 18$ time slices, T simulated shots split evenly over the 19 snapshots, subspace = the support observed in those T shots, 8 independent seeds. Markers are the seed mean; the shaded band and the error bars are ± 1 s.d. over the 8 seeds (at most 8.1% of the mean at any budget, thinner than the line). Upper axis: the sector fraction $|S|/\dim$ reached at that budget; the dotted line is the acceptance threshold $\text{rel-}L_1 = 0.05$ used by the resource scan. At the largest budget measured, $T = 2.60 \times 10^6$ shots, the observed subspace covers 0.835 ± 0.0025 of the sector and the reconstruction reaches $\text{rel-}L_1 = (4.81 \pm 0.27) \times 10^{-3}$. **The shot budget of (a)–(b), measured.** Panel (c) is at another resolution and seed, so the configuration of (a)–(b) was run at finite shots separately: same $\eta = 0.18t$ and $K = 16$, all 8 momenta, both branches, the subspace being the determinants observed in T multinomial draws per channel, 4 seeds, 32 (momentum, seed) pairs, both branches summed in each, scored on the window of (a)–(b); $\pm$ is the s.d. over the pairs. At $T = 5.2 \times 10^6$ shots per channel— 8.3×10^7 over the 16 channels—the reconstruction reaches $\text{rel-}L_1 = (2.16 \pm 0.23) \times 10^{-3}$, maximum 2.46×10^{-3} : panel (b)’s accuracy. It reaches it on a *larger* subspace, covering 0.882 of the addition sector against the 0.850 of (a)–(b), so the budget buys that accuracy together with three extra points of fraction, and not the accuracy at the fraction this figure announces. At half the budget, $T = 2.6 \times 10^6$, the observed subspace covers 0.839—just under the 0.850 of (a)–(b)—and the error is $(6.16 \pm 0.42) \times 10^{-3}$, a factor 2.61 above the mean of (b). That factor, and not the 2.05 transferred from Table S6 ($L = 8$, $\eta = 0.15t$, $|S| = 2180$), is the finite-shot penalty at this configuration; the transferred value understated it by 21% ($1 - 2.05/2.61$).

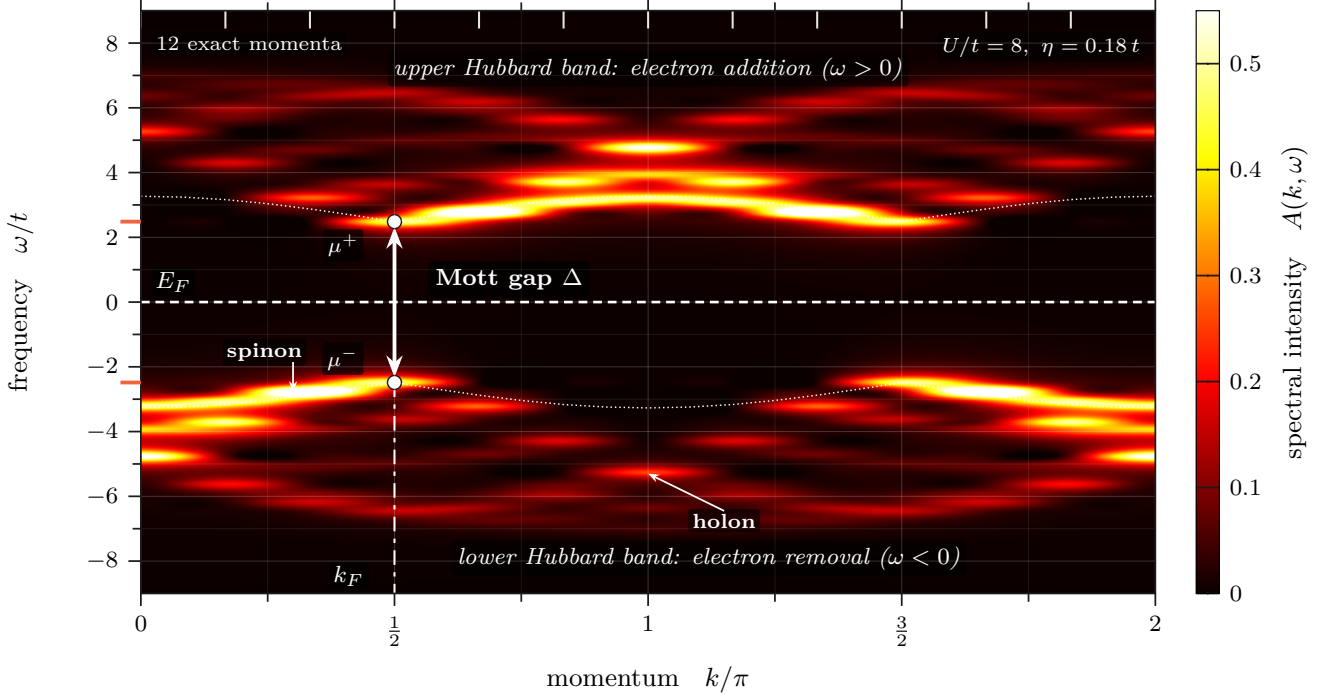


FIG. 4. Momentum-resolved spectral function: the dynamical target. The single-particle spectral function $A(k, \omega)$ of the half-filled periodic Hubbard ring ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18t$), computed from the resolvent $\langle \phi | (z - \hat{H})^{-1} | \phi \rangle$ by a Haydock continued fraction [20] of depth $n_\ell = 150$ over the full $(N \pm 1)$ sectors—a finite-depth continued fraction, that is a 150-node Gauss quadrature of the spectral measure [52] and not the Lehmann sum (Sec. S7)—the frequency- and momentum-resolved observable that sample-based diagonalization targets but its energy-only output cannot itself deliver. The two dispersing Hubbard bands—electron removal (lower, $\omega < 0$; photoemission) and electron addition (upper, $\omega > 0$; inverse photoemission)—are separated by the correlation-driven *Mott gap* $\Delta = \mu^+ - \mu^- = 4.97t$ between the highest occupied state (μ^- , lower edge) and the lowest unoccupied state (μ^+ , upper edge), obtained from the exact $(N \pm 1)$ ground-state energies (ticks at $\pm 2.484t$); the broadened spectral peaks of the plotted $A(k_F, \omega)$ sit slightly wider, at $\pm 2.49t$ (a $4.99t$ splitting), the $\sim 0.02t$ excess being the $\eta = 0.18t$ Lorentzian broadening; the thermodynamic Lieb–Wu value [53] is $4.68t$, the $L = 12$ excess being finite-size. The gap is *direct* at the Fermi momentum $k_F = \pi/2$. Particle–hole symmetry places the Fermi level at mid-gap ($\omega = 0$; here ω is measured from the chemical potential $\mu = U/2$, whereas Figs. 2, 3 and 9 measure it from E_0), verified to double-precision round-off, and each momentum obeys the analytic spectral sum rule $\int A(k, \omega) d\omega = 1$ (the plotted broadened grid recovers it to $\sim 2\%$ within the $\pm 9t$ window). In the thermodynamic limit the lower Hubbard band is spin–charge separated; at $L = 12$ the plot is compatible with a narrow spinon-like branch (bandwidth $\pi J/2 = 0.79t$ [54], $J = 4t^2/U$; dotted, drawn with its mirror image at $\omega > 0$, validated against the exact peak dispersion to $< 0.1t$) that rides a broad holon (charge, $\sim 4t$) band, as expected from the Bethe-ansatz solution [53], and is not offered as evidence of it. At $L = 12$ that structure is three to seven poles per momentum (Table S7), and the in-panel labels “spinon” and “holon” name those branches, not thermodynamic-limit excitations. The momentum axis is a periodic cubic-spline interpolation of the 12 physical momenta (ticks, top; the in-panel “12 exact momenta” counts these allowed momenta, not an exact spectrum). Faithfully *sampling* this full momentum-resolved observable—as opposed to the ground-state energy—requires a large fraction of the $(N+1)$ sector (Fig. 5), the cost quantified in Sec. VII.

A. The objects

Let ϕ be the probe— $\hat{c}_p^\dagger |\Psi_0\rangle$ or $\hat{c}_p |\Psi_0\rangle$ for the charged channels, $\hat{n}_q |\Psi_0\rangle$ or $\hat{S}_q^z |\Psi_0\rangle$ for the neutral ones— H the Hamiltonian restricted to the sector of ϕ , $(N \pm 1)$ or N , and E_0 the ground-state energy of the N -electron sector. Write $z = \omega + E_0 + i\eta$ and $R(z) = (z - H)^{-1}$. The η -broadened spectral function of Eq. (2) is

$$A(\omega) = \frac{\eta}{\pi} \|R(z)\phi\|^2, \quad \text{so that} \quad \int_{\mathbb{R}} A(\omega) d\omega = \|\phi\|^2 \quad (4)$$

for every $\eta > 0$, the sum rule holding because the Poisson kernel integrates to one on each Lehmann pole. Let P be an orthogonal projector on the sector, $Q = I - P$, and

$$\phi_P = P\phi, \quad \phi_Q = Q\phi, \quad w_P = \frac{\|\phi_P\|^2}{\|\phi\|^2}, \quad H_P = PHP, \quad R_P(z) = (z - H_P)^{-1}|_{\text{ran } P}, \quad (5)$$

with Ritz pairs $H_P|\tilde{m}\rangle = \tilde{E}_m|\tilde{m}\rangle$ and $c_m = \langle \tilde{m}|\phi\rangle$. The reconstruction of Eq. (3) is $A_P(\omega) = (\eta/\pi)\|R_P(z)\phi_P\|^2$, and it obeys the same sum rule with $\|\phi_P\|^2$ on the right. When $P = P_S$ is diagonal in the determinant basis—the only case a device produces—we write S for the configuration set and w_S, Λ_S for the quantities below.

The magnitude the theorem is indexed on is the Hamiltonian coupling that crosses the boundary of the subspace, averaged over the resolution window rather than over the spectrum. With $g_m = QH|\tilde{m}\rangle$,

$$\Lambda_P(\eta)^2 := \frac{\eta}{\pi} \int_{\mathbb{R}} \|QH R_P(z)\phi_P\|^2 d\omega = \sum_{m,m'} c_m \bar{c}_{m'} \langle g_{m'}, g_m \rangle \frac{2i\eta}{(\tilde{E}_{m'} - \tilde{E}_m) + 2i\eta}, \quad (6)$$

the second equality being exact and proved in Appendix A; the sum is real, the (m, m') and (m', m) terms being complex conjugates. Two features of Eq. (6) carry the practical content of this paper. It is assembled from objects the reconstruction has already produced—the Ritz vectors and their coefficients—plus one matrix–vector product $H|\tilde{m}\rangle$ per Ritz vector, so evaluating it costs a small multiple of the diagonalization that produced A_P and *does not require* A . And it vanishes exactly when $\text{span}(S)$ contains $\mathcal{K}(H, \phi_P)$, the Krylov space of the retained part of the probe: $QH R_P(z)\phi_P$ vanishes for every z exactly when $\mathcal{K}(H_P, \phi_P)$ is invariant under H , and it then coincides with $\mathcal{K}(H, \phi_P)$. For a subspace that holds the whole probe, $w_P = 1$, the condition is that $\text{span}(S)$ contain $\mathcal{K}(H, \phi)$ itself, which for the local probes here, in the site basis, was measured at $L = 6$ –14 to mean every reflection-allowed determinant of the sector (Sec. V E).

B. The bound

Theorem 1 (a posteriori leakage certificate). *Let $H = H^\dagger$ act on the sector of ϕ , let $\eta > 0$, let P be any orthogonal projector on that sector, and let A, A_P, w_P and $\Lambda_P(\eta)$ be as in Eqs. (4)–(6). Then*

$$\|\phi\|^2(1 - w_P) \leq \|A - A_P\|_1 \leq \min \left\{ \|\phi\|^2(1 + w_P), (\|\phi\| + \|\phi_P\|) \left(\underbrace{\|\phi_Q\|}_{\text{missed weight}} + \underbrace{\Lambda_P(\eta)/\eta}_{\text{boundary leakage}} \right) \right\}, \quad (7)$$

equivalently, dividing by $\|\phi\|^2$ and writing $\hat{\Lambda}_P = \Lambda_P/\|\phi\|$,

$$1 - w_P \leq \frac{\|A - A_P\|_1}{\|\phi\|^2} \leq \min \left\{ 1 + w_P, (1 + \sqrt{w_P})(\sqrt{1 - w_P} + \hat{\Lambda}_P(\eta)/\eta) \right\}.$$

Four hypotheses carry it, each a property of the *pipeline* that produced A_P rather than of the inequality: (H1) A and A_P are referred to the same E_0 ; (H2) the L_1 norm is taken over the whole line and normalized by $\|\phi\|^2$; (H3) A is the exact spectral function of the full sector; (H4) the probe ϕ is built from the exact N -electron ground state. Each is load-bearing and each is checked against this paper’s own pipeline in Sec. S11. A shot-based pipeline breaks (H1), and in general (H4) with it (Sec. III C). If only E_0 is estimated on a sampled subspace, the probe being kept exact, and A_P is referred to $E_0 + \delta$, the bound is guaranteed only once an additive $\|\phi_P\|^2(4/\pi) \arctan(|\delta|/2\eta)$ is added to it, a term the certificate does not control; at $L = 8$, with E_0 from 60% of the N sector, that term moves the bound by +0.20% while the measured error grows by a factor 16 (Sec. S11). The proof, from the Feshbach–Löwdin split of the resolvent (Eq. (A1)) and Cauchy–Schwarz in ω , is in Appendix A.

C. Reading the two branches

What the weight does not control. The first summand of the leakage branch, $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_P}$, is the Born weight of the probe the subspace failed to hold; Table I prices it channel by channel. The second, $\Lambda_P(\eta)/\eta$, reads the off-diagonal block PHQ , and the captured weight does not control it: along the sampler’s ranking it falls as S grows (Sec. V), but a subspace can hold the whole probe and still leak, and Sec. III D measures how much.

A certificate, not an error estimate. The gap between the bound and the measured error is large, and it is not the accident of a constant. At *fixed* P the looseness is stable—within $\pm 17\%$ ($L = 6$) and $\pm 25\%$ ($L = 8$) over a decade in η —but across subspaces it varies by a factor 51, and at the threshold where the leakage branch first improves on the trivial one it is 3×10^2 to 2.4×10^3 (Sec. S4). Equation (7) is therefore an exclusion, not an error estimate; Sec. V reports what it excludes, and at what fraction of the sector.

Neither term is free of the ground state. Given the probe, $\Lambda_P(\eta)$ is computable from H , $\text{ran } P$, ϕ_P and the Ritz vectors alone, without A ; for $P = P_S$, ϕ_P consists of the amplitudes of Ψ_0 on the determinants that the probe operator maps into S , which the reconstruction A_P uses as well. The weight term, $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_P}$, needs the full norm of the probe as well as its retained part ($\|\phi\|^2 = 1 - \langle \hat{n}_p \rangle$ for electron addition). Under (H4) both terms are therefore evaluable without A , but neither without Ψ_0 . If Ψ_0 is itself a sampled-subspace estimate, (H4) fails: the probe is another vector, and the theorem bounds how far the reconstruction lies from that vector’s own spectral function, not from A ; w_P estimated against it is biased in the optimistic direction, and Λ_P is evaluated for the wrong probe, with an error of unknown sign. Nothing the pipeline computes then bounds the distance from its reconstruction to A , and no certificate in this work has been evaluated with such a probe. The complete bound is therefore *not* evaluable by a user who does not already have the ground state. Its regime is the one in which Sec. V evaluates it: exact-diagonalization sizes, where the exact ground state is available and the bound is applied, subspace by subspace, to the configuration sets a sampled pipeline returns; there Sec. VD gives the conservative substitution for w_P that makes the decision itself a posteriori, given the exact probe.

What is new, and the prior art. What this section adds is the conjunction of three things: a closed form for $\Lambda_P(\eta)$ from the Ritz pairs on a subspace of *coordinates*, with no Krylov structure, invariance or gap assumption; a bound two-sided on the same object; and a measured calibration of where it crosses the trivial bound (Sec. VD). Each step of the proof is standard. The split is Löwdin partitioning [55], the algebra of the Feshbach projection [56]; Rayleigh–Ritz and Krylov error analysis, for resolvents and matrix functions [52, 57] and for eigenpairs [28, 58–60], with Paige’s loss-of-orthogonality analysis and the restart convergence theory of Beattie, Embree and Sorensen [61, 62], contains every rung of its steps, and the integration step is the reduced-basis bound on a Galerkin projection with inf-sup constant η [63, 64]. A posteriori estimates for Krylov approximations to matrix functions are due to Frommer and Simoncini, Jia and Lv, and Frommer and Schweitzer [65–67], with Lanczos and stochastic-Lanczos-quadrature versions [68–70]; on quantum Krylov, Oliveira, Ziemas and Glaser assess reliability without knowledge of the true spectrum [71]; and a response-targeted selection criterion for classical selected configuration interaction is Reinholdt, Kjellgren and Kongsted’s [72]. The priority for the a posteriori framing is theirs. At the fractions sampling reaches here the certificate is vacuous (Sec. V, which also measures where it is informative); the full comparison with that literature is in Sec. S2.

D. The captured weight alone does not bound the error

The certificate one would want for a sampled reconstruction indexes the error on the captured weight alone,

$$\|A - A_S\|_1 \leq 2\|\phi\|^2(1 - w_S) + C\rho^{-K_S}, \quad (8)$$

with $w_S = \|P_S\phi\|^2/\|\phi\|^2$ and K_S the largest Krylov order contained in $\text{span}(S)$ (we set $K_S = -1$ when $\phi \notin \text{span}(S)$). It is false, for every constant C and every ρ , and it fails hardest in exactly the regime a sampling experiment aims for.

A two-level counterexample forces the trivial constant. Take $H = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$, $\phi = e_0$, $S = \{0\}$, $E_0 = 0$. Then $P_S\phi = \phi$, so $w_S = 1$ exactly and the weight term vanishes identically; and $\mathcal{K}_0 = \text{span}\{\phi\} \subseteq \text{span}(S)$ while $\mathcal{K}_1 \not\subseteq \text{span}(S)$, so $K_S = 0$ exactly and the second term is the bare constant C . The exact A is a pair of Lorentzians of weight $\frac{1}{2}$ at $\pm g$; the reconstruction, built on $H_S = P_S H P_S = 0$, is one Lorentzian of weight 1 at the origin, so $\|A - A_S\|_1 \rightarrow 2\|\phi\|^2$ as $g/\eta \rightarrow \infty$. Since g is free, Eq. (8) requires $C \geq 2\|\phi\|^2$: false if C is a constant, vacuous if it may depend on the instance. The same holds at every Krylov order: on a chain of $K + 2$ sites with $\phi = e_0$ and S its first $K + 1$ sites, $w_S = 1$ and $K_S = K$, and the exact and reconstructed spectra have strictly interlacing poles, so the same limit requires $C\rho^{-K} \geq 2\|\phi\|^2$ for every K , which no fixed C satisfies when $\rho > 1$. With $w_S < 1$ strictly the weight term alone fails without bound: in a three-level family its violation diverges as ε^{-2} in the amplitude ε of the discarded component (Sec. S9). By contrast $\|A - A_S\|_1 \leq \|\phi\|^2(1 + w_S)$ holds for every S under no hypothesis at all, A and A_S being nonnegative with known integrals: that trivial bound is the only surviving member of the family of Eq. (8), and Theorem 1 keeps it explicitly, as the first branch of its minimum.

The violation is systematic, and worst where the reconstruction is best. In a pooled sweep of 606 subspaces from five Hamiltonian families the weight term of Eq. (8) alone—its $C = 0$ member, $\|A - A_S\|_1 \leq 2\|\phi\|^2(1 - w_S)$, the constant having been disposed of analytically above—fails on 469, and on all 363 with $w_S \geq 0.99$, the mildest by a factor 2.00 (Fig. S5, Sec. S9). In the configuration of Fig. 3 its median violation factor rises from 18.5 to 29.9 between

sector fractions 0.50 and 0.95 while the measured error falls by three orders of magnitude. The sweep and its declared limits are in Sec. S2; the conclusion does not depend on it, the counterexample being analytic.

Why it fails. The reconstruction is Rayleigh–Ritz on $H_S = P_S H P_S$, not a restriction of A to a subset of its eigenpairs: weight the subspace keeps is *displaced* in ω , not discarded, and the two-level example is the extreme case. What the integrals do give is the lower bound $\|\phi\|^2(1 - w_S) \leq \|A - A_S\|_1$, with 0 violations across the 606 subspaces, the left-hand half of Theorem 1.

The two terms are mutually exclusive, and one is off at every operating point here. S is a set of determinants, so P_S is diagonal in the working basis and $P_S \phi = \phi \iff \text{supp}(\phi) \subseteq S \iff w_S = 1$, while $K_S \geq 0$ means precisely $\phi = H^0 \phi \in \text{span}(S)$. Hence the exclusion lemma

$$K_S \geq 0 \iff w_S = 1, \quad (9)$$

verified with 0 discrepancies across the 378 deposited subspaces, of which 357 have $K_S = -1$ (Sec. S9). The two terms of Eq. (8) are therefore never both informative: either $w_S = 1$ and the whole bound is the second term—the two-level counterexample—or $K_S = -1$ and $C\rho^{-K_S} = C\rho$, an undefined constant with no decay in it. Counting settles which branch applies here. The seed is $\phi = \hat{c}_{0\uparrow}^\dagger |\Psi_0\rangle$ at half filling, so $\text{supp}(\phi)$ lies in the set of $(N+1)$ determinants with orbital $0\uparrow$ occupied, a fraction $n_\uparrow/L = \frac{1}{2} + \frac{1}{L}$ of the sector, and fills it except where symmetry zeroes an amplitude of Ψ_0 . By Eq. (9), $K_S \geq 0$ demands $|S| \geq |\text{supp}(\phi)|$, a floor of 0.571 at $L = 8$ (2237 of 3920), 0.600 at $L = 10$, and above 0.54 at $L = 12$ and 14 at any amplitude threshold from 10^{-12} to 10^{-10} , against the sector fractions 0.5561, 0.3625, 0.1700, 0.0800 at which the resource scan of Sec. VII operates (Fig. 5; the coarser replication grid of Table S5 reads 0.180 at $L = 12$ for the same operating point). From $L = 8$ on, every operating point of that scan lies below its own floor, so $K_S = -1$ by counting alone, with no numerics, no ranking and no protocol, and any point below the same line inherits the verdict (Sec. S9). Theorem 1 carries no such bookkeeping: its two terms are defined and simultaneously active for every S , neither is free of Ψ_0 (Sec. III C), and $\Lambda_S(\eta)$ is evaluated, subspace by subspace, in Sec. V.

The magnitude, not the constant. Two measurements at the same, exactly unit, captured weight show that Eq. (8) was indexed not on a bad constant but on the wrong quantity. First, let $S^* = \text{supp}(\phi)$, the smallest determinant subspace of unit captured weight—fixed by the seed, with no ranking, no tie-breaking and no zero-amplitude padding—of size $(\frac{1}{2} + \frac{1}{L})D$ at $L = 6$ and $0.571D$ at $L = 8$, where 426 of the 4900 ground-state amplitudes vanish by symmetry. Its weight term is identically zero, and its relative L_1 error is 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings at $\eta = 0.18t$, rising to 1.15 and 0.73 at $\eta = 0.05t$, against a hard ceiling of $1 + w_S = 2$. Second, S^* is the $K = 0$ rung of the Krylov ladder $S_K = \bigcup_{a \leq K} \text{supp}(H^a \phi)$, every rung of which contains ϕ and so carries $w_S = 1$ identically, while its moment reach grows: S_K reproduces the exact spectral moments through order $2K+1$ —to the 10^{-15} level, and in general failing at $2K+2$, as on the open chain of Table II at both rungs that do not fill the sector—whereas coordinate subspaces of the same size drawn at random miss already μ_0 , in all of 300 draws (Sec. IV). Along a family of constant, exactly unit captured weight the error therefore runs from order unity to moment-exact through order $2K+1$. The captured weight, identical on every rung, does not order them.

What does is elementary. Writing μ_j, μ_j^S for the moments of the unbroadened measures of Sec. IV, every determinant subspace with $w_S = 1$ reproduces μ_0 and μ_1 exactly, and the first moment that can fail is μ_2 , with deficit

$$\mu_2 - \mu_2^S = \|Q_S H P_S \phi\|^2, \quad Q_S = I - P_S, \quad (10)$$

the squared coupling of the Hamiltonian across the boundary of S (E_0 drops out because $Q_S \phi = 0$), zero exactly when S also holds $\text{supp}(H\phi)$ —the $K_S \geq 1$ condition. That boundary coupling is the magnitude the replacement is indexed on: it controls the leakage term through $\Lambda_S(\eta) \leq \|P_S H Q_S\| \|\phi_S\|$, and Theorem 1 averages it over the resolution window rather than over the spectrum. The captured weight says what the subspace *contains*; the error is set by what the Hamiltonian does to the seed at the subspace’s *edge*. Both halves of this paper follow from that displacement: Sec. V measures where the boundary term is small enough for the certificate to say something, and Sec. VI shows why no orbital-rotation invariant can predict it usefully on every state in advance.

IV. MOMENT EXACTNESS ON A KRYLOV-ADEQUATE SUBSPACE

A subspace that contains the order- K Krylov space of the probe reproduces its spectral moments exactly through order $2K+1$ (Theorem 2); the statement is classical, and what is measured here is its exact reach on the coordinate subspaces this work builds. That reach is sharp: on the construction of Table II the moments part company at order $2K+2$, while coordinate subspaces of the same size drawn at random fail already at the zeroth moment. The guarantee is structural: it depends on which determinants the subspace holds, not on how many, and it is decidable from the subspace and the probe, without the exact spectral function. It is not an error bound, and it does not imply convergence in K (Sec. IV B).

TABLE II. **Moment exactness and its exact reach.** Open Hubbard chain, $L = 6$, $U/t = 8$; seed $|\phi\rangle = \hat{c}_{0\uparrow}^\dagger|0\rangle$; $(N+1, S_z = +\frac{1}{2})$ sector of dimension $D = 300$; $\mathcal{S} = \bigcup_{a \leq K} \text{supp}(H^a \phi)$. The largest Krylov order inside $\text{span}(\mathcal{S})$, $K_{\mathcal{S}}$, and the captured Born weight $w_{\mathcal{S}}$ are measured, not assumed. ϵ_j is the *relative* discrepancy of the j -th moment, quoted as the largest value over six equivalent implementations of the same identity (the deposited generator, `src/moments_table.py`, regenerates the roundoff residues to the same order and every other entry to the precision printed); the absolute discrepancy at order $2K+2$ follows in parentheses. Only relative residues are quoted below that order, because μ_j spans five decades between $j = 0$ and $j = 6$ and the absolute roundoff residue is route-dependent (Sec. S10.2). At $K = 2$ the ladder has filled the sector, so $\text{span}(\mathcal{S})$ carries every eigenvector of nonzero Lehmann weight, $A_{\mathcal{S}} = A$ identically, and there is no order- $2K+2$ failure to exhibit; that row is kept to mark where the ladder terminates, and its single entry is the roundoff level of the moments.

K	$ \mathcal{S} $	$ \mathcal{S} /D$	$w_{\mathcal{S}}$	$K_{\mathcal{S}}$	$\max_{j \leq 2K+1} \epsilon_j$	$\longrightarrow$	ϵ_{2K+2} (absolute)
0	200	0.667	1	0	1.8×10^{-15}	$\longrightarrow$	9.5×10^{-4} (3.10×10^{-2})
1	280	0.933	1	1	3.2×10^{-15}	$\longrightarrow$	6.0×10^{-8} (1.34×10^{-4})
2	300	1.000	1	full sector	7.5×10^{-15}	$\longrightarrow$	no failure: $A_{\mathcal{S}} = A$

A. Statement, proof and measured reach

Write $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K\phi\}$ for the order- K Krylov space of the seed $|\phi\rangle = \hat{c}_p^\dagger|0\rangle$, and let $d\mu$ and $d\mu_{\mathcal{S}}$ be the unbroadened Stieltjes measures whose Lorentzian convolutions are the exact A and the reconstruction $A_{\mathcal{S}}$, the Lehmann sum evaluated inside $\text{span}(\mathcal{S})$ as in Sec. II (Sec. S10).

Theorem 2 (Moment exactness). *Let P be the orthogonal projector onto $\text{span}(\mathcal{S})$ and $H_{\mathcal{S}} = PHP$. If $\text{span}(\mathcal{S}) \supseteq \mathcal{K}_K$, then*

$$\int \omega^j d\mu_{\mathcal{S}}(\omega) = \int \omega^j d\mu(\omega) = \langle \phi | (H - E_0)^j | \phi \rangle, \quad 0 \leq j \leq 2K + 1. \quad (11)$$

Proof. Because $\mathcal{K}_K \subseteq \text{span}(\mathcal{S})$ we have $PH^a\phi = H^a\phi$ for $0 \leq a \leq K$, whence by induction $H_{\mathcal{S}}^b\phi = PH^b\phi$ for $0 \leq b \leq K + 1$. Then for any $a \leq K$ and $b \leq K + 1$, using only $P = P^\dagger = P^2$ and $H = H^\dagger$,

$$\langle \phi | H_{\mathcal{S}}^{a+b} | \phi \rangle = \langle H_{\mathcal{S}}^a \phi | H_{\mathcal{S}}^b \phi \rangle = \langle H^a \phi | PH^b \phi \rangle = \langle PH^a \phi | H^b \phi \rangle = \langle \phi | H^{a+b} | \phi \rangle, \quad (12)$$

and every $j \leq 2K + 1$ is such an $a + b$. Both measures are shifted by the same E_0 , so the identity survives term by term in the binomial expansion of $(H - E_0)^j$. $\square$

The proof is three lines of Rayleigh–Ritz, and the statement is the moment-matching property of Krylov projections that underlies Lanczos–Gauss quadrature [52]. The proof does not identify $A_{\mathcal{S}}$ with a Gaussian-quadrature rule, which is not the object built here: every reconstruction diagonalizes $P_{\mathcal{S}}HP_{\mathcal{S}}$ on a coordinate subspace, so $d\mu_{\mathcal{S}}$ carries $|\mathcal{S}|$ nodes, not $K + 1$ (Sec. S3).

Table II measures Eq. (11) on the construction this work uses—the union of the determinant supports of $\{\phi, H\phi, \dots, H^K\phi\}$ —for the open Hubbard chain at $L = 6$, $U/t = 8$, in the $(N+1, S_z = +\frac{1}{2})$ sector of dimension $D = 300$. Through order $2K + 1$ the moments agree to a relative 10^{-14} ; at order $2K+2$ they part company by eleven orders of magnitude at $K = 0$ and seven at $K = 1$. The order $2K + 1$ is the exact reach of the construction, not merely a sufficient claim.

A negative control fixes what the table is about. Over 300 coordinate subspaces of the *same* size $|\mathcal{S}| = 200$ —two thirds of the sector—drawn uniformly from it, every single draw fails already at the zeroth moment, against 4×10^{-16} for the Krylov-support subspace of identical size (Sec. S10.2). What Theorem 2 certifies is *which* determinants the subspace holds, not *how many*.

The ladder terminates once the subspace contains the cyclic subspace generated by ϕ , and there the reconstruction is exact: at $L = 6$ that is $K = 2$ on the open chain, and already $K = 1$ on the periodic rings of Sec. VII, where it stalls at 296 of 300 determinants with $A_{\mathcal{S}} = A$ to 10^{-14} (Sec. S10.2). These three rows are the whole validation, and they say nothing about the large- K behavior of $\|A - A_{\mathcal{K}_K}\|_1$, which is the open question of Conjecture 1.

B. What moment exactness does not buy

Moment exactness is a list of $2K + 2$ linear identities, and the L_1 distance between two Lorentzian-broadened measures is not a finite linear functional of them, so geometric convergence in the Krylov order does not follow from it.

Conjecture 1 (Geometric convergence in the Krylov order). *Fix $\eta > 0$ and let $A_{\mathcal{K}_K}$ be the Rayleigh–Ritz reconstruction on $\mathcal{K}_K$. Are there $C(\eta) < \infty$ and $\rho(\eta) > 1$, uniform over a class of Hamiltonians containing the models studied here, such that $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$ for every K ?*

If the class contains the two-level family of Sec. III D, that family already forces $C(\eta) \geq 2\|\phi\|^2$ at $K = 0$, so any content the conjecture has must be carried by $\rho(\eta)^{-K}$ alone; and the chain family of the same subsection, on which $\text{span}(S) = \mathcal{K}_K$, forces $C(\eta)\rho(\eta)^{-K} \geq 2\|\phi\|^2$ at every K as $g/\eta \rightarrow \infty$, so the class must bound the couplings relative to η . Nor is Eq. (11) an error bound: for a coordinate subspace, containing $\mathcal{K}_K$ forces $w_S = 1$, and at unit captured weight the error is $O(1)$ —0.28 on the open chain of Table II at $\eta = 0.18t$, and up to 0.43 on the periodic rings (Sec. III D). The panel of Fig. 2 read against the conjecture, and the full discussion, are in Sec. S3.

C. The shot count

Proposition 1 counts the shots that populate such a subspace.

Proposition 1 (Sampling capture). *Let $p(x) = \frac{1}{K+1} \sum_{k=0}^K p_k(x)$ with $p_k(x) = |\langle x | e^{-iHt_k} \phi \rangle|^2 / \|\phi\|^2$, draw T bitstrings independently from p , or, for T a multiple of $K+1$, $T/(K+1)$ independently from each p_k as in the protocol of Sec. II, and let $\mathcal{S}$ be the set of distinct outcomes. For $\varepsilon, \delta \in (0, 1)$, if*

$$T \geq \varepsilon^{-1} \log(1/(\varepsilon\delta)) \quad (13)$$

then with probability at least $1 - \delta$ the set $\mathcal{S}$ contains every configuration x with $p(x) \geq \varepsilon$.

The count is written in ε and not in $|\mathcal{S}|$ because the form $T = O(\varepsilon^{-1} \log(|\mathcal{S}|/\delta))$ is circular: $|\mathcal{S}|$ is the random output of the very T draws the bound is meant to fix. With $\mathcal{N}_\varepsilon = \{x : p(x) \geq \varepsilon\}$, each member contributing at least ε to a probability distribution, $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$ *deterministically*; a union bound over $\mathcal{N}_\varepsilon$ of the event “ x never drawn in T trials” gives $\Pr[\mathcal{N}_\varepsilon \not\subseteq \mathcal{S}] \leq \varepsilon^{-1} e^{-\varepsilon T}$, hence Eq. (13). For the stratified draws, $\Pr[x \notin \mathcal{S}] = \prod_k (1 - p_k(x))^{T/(K+1)} \leq e^{-Tp(x)}$, so the same count holds verbatim; and wherever the sampler returns more than ε^{-1} distinct configurations this logarithm is the smaller of the two as well (Sec. S10.4).

The parameter ε thresholds a Born probability, not a spectral accuracy: capturing the high-probability configurations is a statement about the sampler, not about the spectrum (Sec. IV B); what the resulting subspace costs is priced in Sec. VII (Sec. S3 gives the two compositions of the bound this work does not make).

V. WHERE THE LEAKAGE CERTIFICATE IS INFORMATIVE: ITS NON-VACUITY THRESHOLD, MEASURED

Whether the certificate of Theorem 1 says anything on a given subspace is decided by one comparison, computed from the Ritz pairs and the retained probe, given the exact probe (H4). Its *non-vacuity threshold* FR^* is the sector fraction above which the leakage branch of the bound first improves on the trivial bound $\|\phi\|^2(1 + w_S)$, a property of the inequality chain that proves the theorem rather than of the reconstruction. Swept at $L = 6$ –12, the certificate first improves on the trivial bound at a true relative error of 1.2×10^{-3} to 7.3×10^{-3} in every one of sixteen (L, η) cells (Sec. V D); above the threshold it gives converged bounds below the trivial one (Sec. V C); evaluated on the published subspaces at five sizes up to $L = 14$, at the operating points of the resource scan, which lie well below the threshold, it is vacuous (Sec. V B); and the displacement of the threshold is located rung by rung in the proof, with the basis structure behind it (Sec. V E).

Non-vacuity is a condition on one number. With $\hat{\Lambda}_S = \Lambda_S / \|\phi\|$, the leakage branch of Eq. (7) improves on the trivial branch exactly when

$$\frac{\hat{\Lambda}_S(\eta)}{\eta} < \tau(w_S) := \frac{1 + w_S}{1 + \sqrt{w_S}} - \sqrt{1 - w_S}, \quad \tau(1) = 1, \quad (14)$$

so a subspace holding the entire Born weight of the probe certifies something as soon as its leakage, in units of the resolution it was asked for, drops below unity.

TABLE III. **The certificate on the subspaces this paper publishes, including the size the sweep never reached.** Each row is Theorem 1 evaluated at the integer determinant count $|\mathcal{S}| = \text{round}(\text{frac} \times \text{dim})$ that Fig. 5 reports, not at the nearest point of the grid of Table S1. Columns 5–8 are the vacuity factor—the leakage branch of Eq. (7) over the trivial branch $\|\phi\|^2(1 + w_S)$ —so a value above 1 means the certificate says nothing the trivial bound does not. All twenty evaluations are the $T \rightarrow \infty$ limit of the ranking protocol; the sixteen up to $L = 12$ pass the depth test of Sec. VA, and the $L = 14$ row its indicator half only (see below); controls and costs are in Sec. S11. The last column is the measured relative L_1 error at the resolution and against the threshold that *define* the published fraction ($\eta = 0.15t$, $\text{rel-}L_1 < 0.05$), from code sharing neither reference, normalization nor recursion depth with the scan that produced the fractions. At $L = 14$ the ground state is the stored checkpoint, the count is one determinant above the deposited 824 503, and the $K = 18$ ranking, re-run at that size, reproduces the stored selection as a set (Sec. S4). The $L = 14$ row is taken at the deepest recursion run, depth 250, where Λ_{out} was not recorded, so only the indicator half of the depth test applies: the depth indicator is 0.045 and the factor at $\eta = 0.10t$ still falls by 0.17% between depths 200 and 250 ($8.627 \rightarrow 8.613$); the vacuity is not at risk.

L	$\dim(N+1)$	published subspace		vacuity factor at $\eta/t =$				rel- L_1 at $\eta = 0.15t$
		fraction	$ \mathcal{S} $	0.10	0.15	0.18	0.25	
6	300	0.8200	246	4.54	2.94	2.40	1.66	0.0477
8	3 920	0.5561	2 180	5.65	3.69	3.05	2.17	0.0478
10	52 920	0.3625	19 183	6.50	4.28	3.54	2.52	0.0484
12	731 808	0.1700	124 407	8.04	5.20	4.27	3.00	0.0485
14	10 306 296	0.0800	824 504	8.61	5.59	4.61	3.26	0.0463

A. What is evaluated

We evaluate Theorem 1 on the subspaces of Fig. 5 at $L = 6, 8, 10$ and 12 , with the ranking protocol of Sec. VII, over fractions 0.20 – 0.95 and $\eta = 0.10$ – $0.25t$: 204 evaluations, of which 187 are converged in Krylov depth by the depth test of the sweep—the internal indicator $\Lambda_{\text{in}}/\Lambda_{\text{out}}$, the truncation part of Λ over its boundary part, below 0.05 , and Λ_{out} constant to 10^{-3} relative over the last three depths of the ladder—and no row flagged not converged is quoted. To that grid we add twenty evaluations at the published subspaces themselves, at the integer counts 246, 2180, 19 183, 124 407 and 824 504 at $L = 6$ – 14 . The sixteen up to $L = 12$ pass the depth test and the $L = 14$ row its indicator half only (Table III). The subspaces are the $T \rightarrow \infty$ limit of the protocol, not finite-shot draws. What is certified is the object the code forms, the reorthogonalized Krylov reconstruction A_K at a declared depth, against a reference whose drift is declared, and no measured error within a factor 20 of that drift is quoted. The protocol, the controls and the costs are in Secs. S4 and S11.

B. The certificate at the operating fractions of the resource scan

Evaluated on the published subspaces themselves, the leakage branch of the bound exceeds the trivial bound $\|\phi\|^2(1 + w_S)$ at all five sizes and all four resolutions, by factors running from 1.66 ($L = 6$, $\eta = 0.25t$) to 8.61 ($L = 14$, $\eta = 0.10t$), and by 2.40 to 4.61 at the $\eta = 0.18t$ of the momentum-resolved channel; Table III carries the twenty values. They are monotone in both arguments—rising with L at every resolution, falling with η at every size—so the certificate is furthest from non-vacuity at the largest sector and the finest resolution, the regime the method targets.

The largest classically converged case carries the evidence. At $L = 12$ and the published subspace— $|\mathcal{S}| = 124 407$ of 731 808 determinants— $w_S = 0.998622$, so Eq. (14) asks for $\hat{\Lambda}_S/\eta < \tau(w_S) = 0.9625$; the measured value is 4.236 at $\eta = 0.18t$, converged in depth (Sec. S4). Over the 606 live subspaces of the pooled sweep the leakage branch is the smaller of the two on only 61. The reconstructions at the operating fractions of the resource scan—Fig. 5 and the operating points of Secs. VII and VIII—are therefore uncertified; the Born-ranked reconstruction of Fig. 3, at 85% of its sector, lies above the threshold (Sec. VC).

Two controls come with the table. Applying this paper’s own acceptance criterion to the same sweep with that independent code returns the reported fractions to under 2% (Table S1), and the last column of Table III evaluates the defining criterion at the published subspaces themselves, 0.0463–0.0485 against the threshold 0.05. The reconstructions and the curve are thus reproduced independently; what those fractions lack is a certificate.

C. What the certificate certifies, and at what shot cost

What the sweep does certify, at each size the strongest statement whose bracket is converged, is collected in Table S3: at $L = 12$, 60% of a 731 808-dimensional sector gives relative $L_1 \leq 0.717$, a factor 2.8 below the trivial cap, tightening to ≤ 0.218 at 70% and $\eta = 0.25t$. One further point of the map lies above the threshold: for the Born-ranked (infinite-shot) reconstruction of Fig. 3, at 85% of its sector, the certificate of the depth-500 Krylov reconstruction is non-vacuous in all sixteen channels, at a relative bound of 0.77–1.75 against the trivial 2, and loose, 350–900 times the true error of the channel integrated over the whole frequency line (`cert_akw.json`). At the acceptance criterion used throughout this paper, relative $L_1 < 0.05$, which the reconstruction at $L = 8$, $\eta = 0.18t$ reaches at $\text{FR} \approx 0.52$, the certificate requires at least 0.98 of the sector at $L = 8$, the crossing lying in $[0.977, 0.983]$, and on our grid it is not reached at all at $L = 6, 10$ or 12 : at that accuracy Theorem 1 does not certify the operating points of Fig. 5, and the regime where it would costs more shots than the method saves, since reaching the non-vacuous band multiplies the shots per snapshot needed for the marginal determinant by 7.5–10 at $L = 8$ and by 20–35 at $L = 12$ (Table S2).

D. The crossing of the trivial bound is calibrated

Across all sixteen (L, η) cells the crossing of the trivial bound occurs at a true relative L_1 error of 1.2×10^{-3} to 7.3×10^{-3} , median 2.8×10^{-3} : a spread of 5.9 over true errors spanning five decades, and in every cell at least 47 times the drift of our own reference. The spread is a drift, not scatter, and it runs the conservative way in L : the larger the system, the smaller the true error at which the certificate first says anything. The crossing of $\tilde{\Lambda}_S(\eta)/\eta$ therefore locates a true error to within that factor, and it is computed from H , S , the retained probe ϕ_S , the probe norm $\|\phi\|$ and η , without the spectral function it bounds, given the exact probe (H4), with four limits (Sec. S4). It locates one error level per cell and is not an error estimate. A user without w_S can discharge $\tau(w_S)$ conservatively: an assumed floor $w_S \geq 0.995$ gives $\tau \geq 0.9280$, moving the threshold by at most 7%. The calibration rests on these sixteen cells, $L = 6$ – 12 , and is not extrapolated to $L = 14$. A two-constant rule in $1 - w_S$, $1 - w_S < c\eta^p$ with c and p fitted on some sizes and tested on the others, localizes the crossing error more tightly than the certificate on eleven of the fourteen train/test splits (`null_ws7.py`): as a tracker of the error the certificate has no edge in sharpness over the captured weight. Its edge is provability. It is an inequality with a proof, whose decision survives replacing w_S by a floor, whereas the fitted rule is a regularity of these systems with no guarantee outside its fitted range—no inequality indexed on the captured weight alone bounds the error (Sec. III D)—and it asks for $1 - w_S$ to 10^{-5} – 10^{-4} and, stripped of its second parameter, is the looser of the two on thirteen of the fourteen splits.

E. Where the threshold comes from

Measured rung by rung along the chain of inequalities that proves Theorem 1, at $L = 8$, the displacement of the threshold is 44% the pointwise step to a Hilbert-space distance, 38% the operator-norm step $\|R(z)\| \leq 1/\eta$, 11% the Feshbach/Minkowski split and 4.6% Cauchy–Schwarz in ω , whose total margin for tightening is a factor 1.05 typically and 3.29 at worst. The subspace adds one structural fact. For a subspace holding the whole probe, $\Lambda_S = 0$ requires $\text{span}(S) \supseteq \mathcal{K}(H, \phi)$, and the reflection $j \rightarrow -j$ about the seed site commutes with H and has ϕ as an eigenvector, so every vector of $\mathcal{K}$ vanishes on the reflection-fixed determinants of the opposite parity—4, 18, 48, 200 and 600 at $L = 6$ – 14 —while a reflection-projected power iteration (`z_suppK_sym.py`) finds every other determinant in the support: 296, 3902, 52 872, 731 608 and 10 305 696 at $L = 6$ – 14 . At full captured weight, therefore, no coordinate subspace short of all of them has zero leakage. That is a statement that $\Lambda_S \neq 0$, not about its size; the vacuity at the operating fractions of the resource scan, all below full captured weight, is measured in Table III, not deduced from it. Nor is the ranking the weak link: a Krylov-diagonal oracle ranking gives a *worse* certificate than the Born ranking at $L = 6$, $\eta = 0.18t$, at every fraction checked from 0.50 on. At full captured weight the leakage branch is of first order in the boundary coupling and the error of second, and the same identity gives an a posteriori certificate of second order,

$$\|A - A_P\|_1 \leq \frac{\Lambda_P(\eta) \tilde{\Lambda}_P(\eta)}{\eta^2} \quad (\phi_Q = 0), \quad (15)$$

where $\tilde{\Lambda}_P$ is Eq. (6) with the energy difference in its kernel reversed in sign (equal to Λ_P for a real Hamiltonian and probe) (Sec. S4); whether one exists below full captured weight, where every published fraction lies, is open. The measurement basis is a choice, and for the momentum-resolved probe of Fig. 3 it is a forced one: the Krylov space lies inside one momentum block of exactly $\dim(N+1)/L$ plane-wave determinants, on which the certificate becomes non-vacuous at 48–50 of 50 determinants ($L = 6$) and 455–483 of 490 ($L = 8$) (`z_kblock.py`), a factor 5–7 fewer than

the site determinant basis requires. The block exists only after the measurement is rotated to the plane-wave orbitals, about $L(L-1)$ further two-qubit gates whose effect on the sampled distribution is unmeasured (Sec. S4); in the site basis the same probes at $L=8$ have support on 3848 to 3920 of the 3920 determinants (Sec. S2). The gain is $1/L$ against an exponential, and whether the support fills the block at $L \geq 10$ is unmeasured.

VI. NO SHORTCUT TO THE BOUNDARY: AN OBSTRUCTION AND ITS WITNESS

Section III leaves one question open: can the a posteriori certificate be replaced by an a priori predictor computed from the state before any circuit runs? The natural candidates are the monotones of fermionic non-Gaussianity, efficiently measurable from two-point functions (Sec. S5 reviews them). None can play that role, for either term of Eq. (7): not for the static determinant support $|\mathcal{S}|_{\varepsilon_w}$ of the ground state that the shots must populate, and not for the leakage $\Lambda_S(\eta)/\eta$ of the coordinate subspace the sampler returned. What replaces the predictor is the measured map of Sec. V. Three results establish this. On number-conserving states the one-body antiflatness is a single invariant of the one-particle density matrix (Eq. (17)); a geminal family carries antiflatness of every order at bond dimension 2 (Theorem 3); and on fibers on which every invariant is frozen the subspace-aware certificate still separates, so the leakage bound that invariants do supply is vacuous there (Proposition 2).

A. The collapse: one-body magic is a single 1-RDM invariant

We work with the fermionic AntiFlatness (FAF) [16],

$$\mathcal{F}_k = 2n_o - \frac{1}{2} \text{tr}[(\Gamma^\top \Gamma)^k] = 2n_o - \sum_{j=1}^{2n_o} \lambda_j^{2k}, \quad \Gamma_{mn} = -\frac{i}{2} \langle [\gamma_m, \gamma_n] \rangle, \quad (16)$$

with Γ the $4n_o \times 4n_o$ Majorana covariance matrix, $\{\lambda_j\} \in [0, 1]$ the moduli of its eigenvalue pairs $\pm i\lambda_j$, and $2n_o$ the number of spin-orbital *modes* (twice the spatial-orbital count), distinct from the electron number of the $(N \pm 1)$ sectors. Each $\mathcal{F}_k$ vanishes if and only if the state is Gaussian and is invariant under Gaussian (free-fermion) unitaries by construction [16]; the index k is the replica (moment) order of $\text{tr}[(\Gamma^\top \Gamma)^k]$, and every $\mathcal{F}_k$ is efficiently measurable from two-point functions [16, 17, 73].

On the number-conserving states of electronic structure the anomalous correlators $\langle c_i c_j \rangle$ vanish, so Γ is fixed by the one-particle reduced density matrix γ and its singular values are $\lambda_i = |2g_i - 1|$, with $g_i \in [0, 1]$ the natural spin-orbital occupations. The $k=1$ member then reduces to a single 1-RDM invariant,

$$\boxed{\mathcal{F}_1 = \sum_i [1 - (2g_i - 1)^2] = 4 \sum_i g_i(1 - g_i) = 4 \text{tr}[\gamma(1 - \gamma)]}, \quad (17)$$

exact on any number-conserving state and verified state by state to 10^{-13} on the 38 molecular ground states of this work. For spin-restricted natural orbitals it equals $2N_u$, the effectively-unpaired-electron index [74, 75], and $\text{tr}[\gamma(1 - \gamma)]$ is one quarter of the one-body quadratic (linear) entropy [76], a trace-form one-body entanglement entropy [77]. The algebra is an immediate number-conserving specialization of the antiflatness of Ref. [16], noted independently for fixed-particle-number states in a study of interacting integrable models [78]. Being a function of the occupation spectrum alone, $\mathcal{F}_1$ is invariant under orbital rotations (the spin-polarized cases and the dictionary are in Sec. S5).

B. The geminal witness, and what an invariant cannot certify

What obstructs the operative quantities—the thresholded support $|\mathcal{S}|_{\varepsilon_w}$ and the leakage—is a symmetry, exhibited by a family whose amplitudes are $\Theta(1)$ and therefore threshold-robust: an antisymmetrized product of K strongly orthogonal singlet geminals (perfect-pairing / generalized-valence-bond),

$$|\Psi_K\rangle = \bigotimes_{i=1}^K \left(\cos \theta \hat{b}_{i\uparrow}^\dagger \hat{b}_{i\downarrow}^\dagger + \sin \theta \hat{a}_{i\uparrow}^\dagger \hat{a}_{i\downarrow}^\dagger \right) |0\rangle, \quad (18)$$

on $2K$ spatial orbitals ($4K$ modes, $N = 2K$ electrons), each geminal occupying a disjoint bonding/antibonding pair (b_i, a_i) .

Theorem 3 (Geminal orbit witness). *At maximal pairing $\theta = \pi/4$ the family (18) satisfies, exactly,*

$$\mathcal{F}_1 = 2N_u = 4K, \quad \mathcal{F}_2 = 4K, \quad |\mathcal{S}|_{\text{pair}} = 2^K, \quad \chi = 2, \quad (19)$$

with $\mathcal{F}_{1,2}$ the order-1 and order-2 antifatness of Eq. (16), $|\mathcal{S}|_{\text{pair}}$ the determinant support in the pairing basis, and χ the minimal matrix-product-state bond dimension in the pairing ordering. In that localized (pairing) basis the state is a bond-dimension-2 tensor network, representable, contractible and Born-samplable in $O(K)$.

One-body magic, higher-order magic and determinant support therefore all diverge with K : magic of every order is unbounded at bond dimension 2, where the simulation cost grows only linearly, as $O(K)$. The proof, four block-diagonal computations, is given in the Supplemental Material (Sec. S12). The values of $\mathcal{F}_1$, $\mathcal{F}_2$ and $|\mathcal{S}|_{\text{pair}}$ are reproduced by exact diagonalization for $K = 1, \dots, 4$, and so is $\chi = 2$, the largest Schmidt rank over all cuts of the pairing ordering (Fig. S1 and Sec. S12). The orbit is the argument: $\mathcal{F}_k$ is invariant under number-conserving orbital rotations [16], so the whole orbit $\{U|\Psi_K\rangle\}$ carries the same $(\mathcal{F}_1, \mathcal{F}_2) = (4K, 4K)$ while the cost in the fixed computational basis is $O(K)$ at the pairing point and, at generic rotated points, not established here (Born sampling there is plausibly hard, Sec. S5). A Gaussian-invariant fermionic magic therefore does not lower-bound the bond dimension, nor the cost beyond linear order in K , and it determines the cost only if the plausibly hard rotated points are in fact easy. The same orbit reaches the second budget of Eq. (7):

Proposition 2 (Fibers on which every orbital-rotation invariant gives a vacuous leakage certificate). *Let $\mathcal{U}_U$ be the Gaussian unitary implementing $U \in \mathcal{U}(n_o)$, and let the same physical problem written in the rotated orbital basis be $(H_U, \Psi_U, \phi_U) = (\mathcal{U}_U^\dagger H \mathcal{U}_U, \mathcal{U}_U^\dagger \Psi, \mathcal{U}_U^\dagger \phi)$, the determinant basis—in which the device measures, and of which S is a coordinate subspace—held fixed. Along this orbit the spectrum of H , E_0 , $\|\phi\|^2$, every Lehmann weight and hence $A(\omega)$ pointwise are constant, as is every orbital-rotation invariant of the state, in particular $\mathcal{F}_1 = 4\text{tr}[\gamma(1-\gamma)]$ and all $\mathcal{F}_k$. $\Lambda_S(\eta)/\eta$ is not. An upper bound on $\Lambda_S(\eta)/\eta$ is available from invariants alone—combining $\Lambda_S \leq \|P_S H Q_S\| \|\phi_S\|$ with $\|Q_S H P_S\| = \|Q_S(H - c)P_S\| \leq \min_c \|H - c\|$ gives $\Lambda_S(\eta)/\eta \leq (W/2)\|\phi\|\sqrt{w_S}/\eta$, W the spectral width of the $(N \pm 1)$ sector—but it is vacuous by a factor of at least $W/2\eta$, which grows extensively with system size. That no invariant bound can be useful on these fibers is seen at fixed invariants: any $B(\text{invariants}, |\mathcal{S}|, w_S)$ valid for $\Lambda_S(\eta)/\eta$ is constant on a fiber of fixed arguments and must therefore exceed the fiber maximum. In the geminal family APSG($\pi/4$) on $2K$ orbitals with $H = \varepsilon \hat{N} - g \sum_i A_i^\dagger A_i + \delta h$ we exhibit fibers ($K = 2, 3, 4$; $\delta = 0.01$; $\eta = 0.10$; $|\mathcal{S}| = 2^{K-1}$; spectrum, Lehmann weights and all $\mathcal{F}_k$ matched to 10^{-14} , w_S to 2.2×10^{-16}) whose endpoints carry $\hat{\Lambda}_S(\eta)/\eta = 0.237$ and 2.326 , 0.323 and 2.336 , 0.391 and 2.346 . Since the larger value exceeds $\tau(w_S) = 0.795$ of Eq. (14), above which the leakage branch of Theorem 1 no longer improves on the trivial bound $\|\phi\|^2(1 + w_S)$, every valid invariant bound is vacuous on these fibers—while the subspace-aware certificate is non-vacuous at the aligned endpoint (0.856 against a trivial 1.962). The statement is one of non-determination and non-vacuity, not of non-existence: invariants do not fail to bound the leakage, they fail to bound it usefully.*

The fibers (Table S4), an exhaustive search over the 276 subspaces of that size at $K = 2$, and four limits of the exhibit—a matched pair at one angle, a margin over the trivial bound that shrinks from 2.29 to 1.69 between $K = 2$ and $K = 4$, a small-leakage endpoint realized only by the stabilizer of the state, and sector dimensions of at most 3920—are in Sec. S5.

The raw determinant support—the unthresholded Slater rank—is obstructed separately, by continuity: no continuous functional of a fixed set of reduced density matrices can upper-bound the count more tightly than the sector dimension near a single determinant, states arbitrarily close to one in every such functional carrying any support up to the whole sector, rank being coefficient-insensitive and not upper-semicontinuous (though, unlike the CP tensor rank, whose sets of rank at most r are not closed [79], it is lower-semicontinuous); and at half filling the invariant $\mathcal{F}_1$ cannot lower-bound the count beyond 2, the two-determinant cat having maximal $\mathcal{F}_1$. That continuity argument (Sec. S13) does *not* transfer to the operative, coefficient-sensitive $|\mathcal{S}|_{\varepsilon_w}$, which collapses to 1 along the same path as the functionals; the symmetry above does.

The cost side: a bilateral obstruction whose sign is not stable. Unlike $\mathcal{F}_1$, the determinant support is basis dependent, and both directions of the decoupling are realized (Fig. S2): a free-fermion ground state has $\mathcal{F}_1 = 0$ and a site-basis support $|\mathcal{S}|_{\varepsilon_w}$ (open chain, $\varepsilon_w = 2.5 \times 10^{-3}$) of 35, 336, 3496 and 37361 at $L = 4$ –10; along a Hubbard chain raising U raises $\mathcal{F}_1$ while lowering $|\mathcal{S}|_{\varepsilon_w}$; across the nineteen molecules the two rise together. Within each of the six (L, N) strata of the Hubbard sweep the rank correlation of $\mathcal{F}_1$ with $|\mathcal{S}|_{\varepsilon_w}$ is exactly -1 (stratified Kendall $\tau = -1.000$, exact one-sided permutation $p = 3.35 \times 10^{-13}$ against independent orderings within each stratum, a null that six strata sharing one mechanism do not satisfy), while it is $+1.000$ in the chain and ladder families and $+0.86$ across the molecules: an excellent rank predictor whose sign reverses between families, as an orbital invariant compared with a basis-dependent count must. The bond dimension χ , a rigorous floor on the raw support, tracks the thresholded one

better (mean within-stratum $\rho = 0.948$), but it is itself basis dependent, measured rather than predicted, and blind to the leakage. Computable Gaussian-invariant measures of non-Gaussianity include the antiflatness used here [16] and the two families of Tarabunga *et al.* [17], whose natural-orbital participation entropies, themselves Gaussian invariant, also upper-bound the classical simulation cost in an orthonormal Gaussian basis; their quadratic occupation entropy, which equals $\mathcal{F}_1$, is a strong pure-state monotone, proved by Tarabunga *et al.* and, independently, by Leone and Bittel [18]. The contrast between an orbital invariant and a basis-dependent count is our observation, and what the construction adds is a family on which every order of the invariant is maximal at $O(K)$ cost, and fibers on which the invariants are frozen while the certificate of Theorem 1 still separates. The stratified statistics, the molecular set, which does not discriminate, and the two families where $\mathcal{F}_1$ wins are in Secs. S5 and S14.

VII. THREE PRICES: SUPPORT, RESOLUTION, AND SHOTS

On half-filled periodic Hubbard rings at $U/t = 8$ ($L = 4$ –14, 8 to 28 qubits) the method is priced on three axes, each measured here: the determinant support the sampler must populate grows as $e^{1.05L}$, more slowly than the sector's $e^{1.30L}$; the sector fraction, unlike the support, is a property of the question put to the model rather than of the model, its exponent moving by a factor 2.2 over a decade of the resolution η ; and the shot budget T that populates the support, measured at $L = 4$ –12, lies two orders of magnitude above it and grows as $e^{1.12L}$ over $L = 6$ –12, its $L = 14$ value being a floor of 1.3×10^8 shots for a single seed and branch.

The dynamical support. $|\mathcal{S}|_\eta$ here is the number of determinants needed for the reconstruction of $A(\omega)$ to reach relative $L_1 < \text{thr}$ at resolution η and recursion depth n_ℓ , so it depends on all three (`scaling_lanczos_mf.py`, `stage_frac`: `thr = 0.05`, $\eta = 0.15t$, $n_\ell = 150$). It is distinct from the static ground-state support $|\mathcal{S}|_{\varepsilon_w}$ of Sec. VI and from the returned set S of Theorem 1: the three coincide only by accident, and no statement here transfers to the other two (Sec. I).

A. The support exponent and its resolution dependence

Across $L = 4$ –14 (8 to 28 qubits, half-filled *periodic* rings at $U/t = 8$; the three largest sizes reached by matrix-free Lanczos/Haydock and validated against dense diagonalization at $L \leq 8$), the determinant support needed to reconstruct $A(\omega)$ to relative $L_1 < 0.05$ at $\eta = 0.15t$ and $n_\ell = 150$ grows exponentially,

$$|\mathcal{S}|_{\eta=0.15} = 22, 246, 2180, 19183, 124407, 824503 \implies |\mathcal{S}|_\eta \sim e^{1.05L} \quad (L = 4\text{--}14, R^2 = 0.998), \quad (20)$$

more slowly than the $(N+1)$ sector itself, $\dim = 24, 300, 3920, 52920, 731808, 10306296 \sim e^{1.30L}$. Restricted to $L = 4$ –12, the five sizes of the replication grid, the fit gives 1.082 at $R^2 = 0.998$. The fraction is the support divided by the sector dimension, exactly, so the fraction and support exponents differ by the constant $b_{\dim} = 1.291$ and move together, by 0.148 over the decade $\eta = 0.05 \rightarrow 0.50t$ (Table S5): a seventh of the support exponent, three quarters of the fraction exponent. We therefore report the support, the absolute resource a device has to populate, and the fraction as the derived, resolution-dependent quantity that it is.

B. The sector fraction is a property of the question, not of the model

At $\eta = 0.15t$ the fraction needed for a fixed spectral accuracy falls from 0.92 at $L = 4$ to 0.08 at $L = 14$, but its exponent is set by η : over the decade of η it moves by a factor 2.2, from -0.1266 to -0.2743 (Table S5), and even at fixed η the local logarithmic slopes between consecutive published points steepen by a factor 6.8 across five intervals, so the fraction is not a single exponential either. The reference is itself a depth-150 Lanczos computation whose drift between depths 100 and 200 at $L = 12$ is 0.93% at $\eta = 0.15t$ and 3.0% at $\eta = 0.10t$, so the fraction is reported as a declared window, $\eta \geq 0.15t$ at $n_\ell = 150$, with the full grid and the finite-depth caveats in Sec. S6.

C. The third price: shots

Populating a subspace by measurement is a coupon-collector problem on the Born distribution of the time-evolved seed, so the budget exceeds the support it buys. Table IV reports that budget *measured*, by multinomial sampling of

TABLE IV. **The third price, measured at every size but the largest.** T is the total shot budget, split evenly over the 19 snapshots of the protocol, for a single orbital seed and a single branch; it is the budget at which the finite-shot reconstruction itself first reaches relative $L_1 < 0.05$, and $\pm$ is the sample standard deviation across seeds, not the standard error of the mean. Run A is `sampled_honest.json` (8 seeds, $L \leq 8$); run B is `honest_sampling.json` (4 seeds, 6 at $L = 10$), and the two are separate runs with different seeds and different budget grids, listed separately at $L = 6$ and $L = 8$. At $L = 4$ run B is censored by its own cap of 10^8 shots per snapshot, the 24-state sector being degenerate for this purpose, and run A is quoted there. The last column divides T by the *published* support; dividing instead by the support each run actually populated gives 95, 107, 141 and 124 at $L = 6$ –12. Only $L = 14$ is a floor: the overhead held at its $L = 12$ value, which is a lower bound and not an estimate.

L	$\dim(N+1)$	$ \mathcal{S} _\eta$	T	$T/ \mathcal{S} _\eta$	status
4	24	22	$(7.8 \pm 2.0) \times 10^2$	35	measured, run A, 8 seeds
6	300	246	$(2.43 \pm 0.34) \times 10^4$	99	measured, run A, 8 seeds
6	300	246	$(2.44 \pm 0.23) \times 10^4$	99	measured, run B, 4 seeds
8	3920	2180	$(2.70 \pm 0.13) \times 10^5$	124	measured, run A, 8 seeds
8	3920	2180	$(2.61 \pm 0.08) \times 10^5$	120	measured, run B, 4 seeds
10	52920	19183	$(3.09 \pm 0.04) \times 10^6$	161	measured, run B, 6 seeds
12	731808	124407	$(1.91 \pm 0.01) \times 10^7$	154	measured, run B, 4 seeds
14	10306296	824503	$\gtrsim 1.3 \times 10^8$	154	floor, not measured

the declared protocol, at $L = 4$ –12, in two independent runs that agree where they overlap to 0.2% and 3.2%; every budget is for *one* orbital seed and *one* branch.

The budget is two orders of magnitude larger than the support it buys, already at $L = 6$, and it grows exponentially, 7.8×10^2 to 1.9×10^7 over $L = 4$ –12, a log-linear fit over $L = 6$ –12 giving $T \sim e^{1.12L}$ at $R^2 = 0.996$. At $L = 8$ the sixteen-channel momentum-resolved configuration of Fig. 3 reached its own resolution and accuracy with 8.3×10^7 , and not with half as many; the $16 \times 2.7 \times 10^5 \approx 4 \times 10^6$ shots that sixteen channels at the local-seed budget would cost is a transfer and not a measurement, because the momentum seeds have larger support. The $L = 14$ floor of 1.3×10^8 shots for a *single* seed and branch (Table IV) exceeds the largest shot budget drawn anywhere in this work, the 8.3×10^7 shots of the sixteen-channel configuration of Fig. 3. The overhead per determinant is a level and not a growth law—35, 99, 124, 161, 154 against the published support across $L = 4$ –12, rising by a factor 3.5 over the first three sizes (35 to 124), then to 161 at $L = 10$, and not rising over the last step, 161 to 154. Its exponent is not resolved by these data. Fitted on $L = 6$ –12 it is 0.08 per site against the published support (95% interval $[-0.03, 0.19]$) and 0.054 ($[-0.066, 0.174]$) against the support each run populated; both intervals contain zero, so *these data do not establish that the shot budget grows with a larger exponent than the support*, only that it is larger in absolute terms at every size measured.

What the published curve is. The published curve is the $T \rightarrow \infty$ limit of the declared protocol: the ranking that produced the published fractions is `argsort` applied to the Born distribution of the time-evolved state accumulated over the 19 snapshots of the protocol—the distribution the device would itself measure with infinitely many shots—and not a clairvoyant ordering by exact eigenvector amplitudes. With finite shots the fraction needed rises, from 0.8200 to 0.8571 ± 0.0135 at $L = 6$ and from 0.5561 to 0.6275 ± 0.0031 at $L = 8$ (Table S6), so the published curve is a strict lower bound on the finite-shot cost. At fixed size the penalty is budgetary, not structural: at $L = 8$ raising the budget by a factor 12.7 takes the ratio of the finite-shot error to the infinite-shot-ordering error from 1.96 ± 0.12 to 1.06 ± 0.01 . Finite-shot fractions are therefore reported only with their budget and seed count attached.

VIII. PHYSICAL VALIDATION: THE TWO GAPS SCALE IN OPPOSITE DIRECTIONS

Across $L = 6$ –12 the exact full-sector references against which the sampled reconstructions are scored reproduce the opposite finite-size scaling of the two collective scales of the half-filled Hubbard ring. On the same lattice, the charge gap decreases monotonically to within 6.2% of its Lieb–Wu thermodynamic value, while the spin gap closes as $\approx 2/L$, with an independent effective-Heisenberg control (Table V). The answer is known exactly and independently of this work, so the scaling, reported across four sizes rather than as a picture at a single size, is a check of the references and not a result (Sec. VIII C).

A. Charge converges to a finite value, spin closes as $1/L$

Charge. $\Delta(L)$ decreases monotonically and stays above the Lieb–Wu value $4.67952t$ at every size, the residual falling from $0.679t$ at $L = 6$ to $0.289t$ at $L = 12$, 6.2% of the thermodynamic value; $\Delta(L = 12) = 4.96876t$ is the

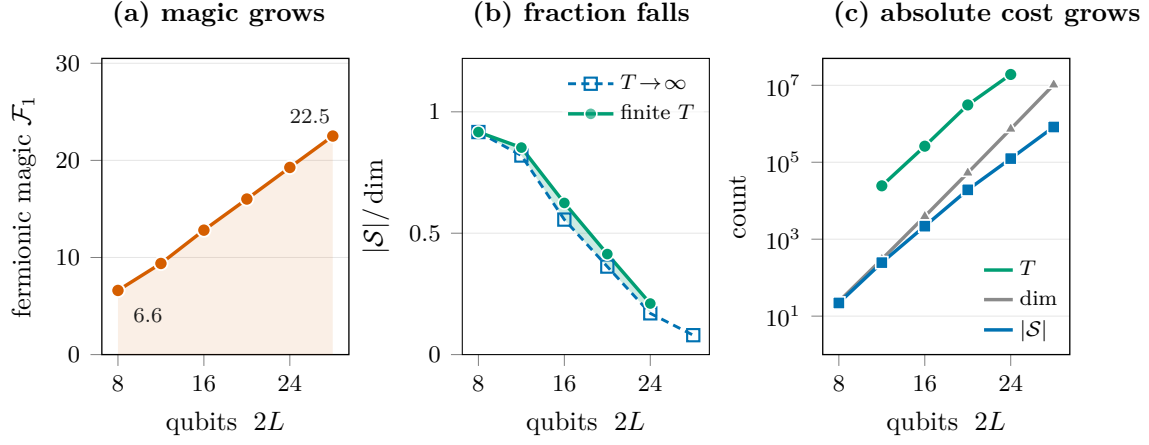


FIG. 5. The resource scan: one-body magic, the required sector fraction and the absolute counts. One-dimensional Hubbard ring, $U/t = 8$, half filling, periodic boundary conditions; the subspace is built in the $(N+1, S_z = +\frac{1}{2})$ sector seeded by $c_{0\uparrow}^\dagger |\Psi_0\rangle$ from the $K+1 = 19$ time snapshots of the declared protocol ($K = 18$, $\delta t = 0.5$), and a subspace is accepted when its Haydock reconstruction reaches relative- $L_1 < 0.05$ against the exact spectrum at broadening $\eta = 0.15t$. All fractions below are at that single η . **(a)** One-body fermionic magic of the ground state, 6.6074 at $L = 4$ to 22.5050 at $L = 14$ (periodic ring; the open-chain families elsewhere in this paper carry different values and must not be substituted here). It is not a cost: its rank correlation with the support reverses sign between families (Sec. VI). **(b)** The required fraction of the sector, twice. Open squares, dashed: the infinite-shot limit of the same protocol — the subspace obtained by ranking the exact Born weights of the snapshots — 0.9167, 0.8200, 0.5561, 0.3625, 0.1700, 0.0800 for $L = 4$ –14, the published fractions quoted throughout this paper. Filled circles, solid: the same acceptance criterion reached by actually drawing shots (classically simulated multinomial draws from the exact time-evolved state), 0.9167 ± 0.0000 , 0.8527 ± 0.0096 , 0.6246 ± 0.0033 , 0.4141 ± 0.0012 , 0.2104 ± 0.0004 (mean $\pm$ s.d. over 4 independent shot seeds; 6 seeds at $L = 10$); the error bars are drawn and are smaller than the markers. Finite sampling needs +4.0%, +12.3%, +14.2% and +23.7% more of the sector than the infinite-shot limit at $L = 6, 8, 10, 12$: the penalty does not cancel, it grows with size, and the shaded strip is that difference. An independent 8-seed run of a separately written sampler reproduces 0.8571 ± 0.0135 ($L = 6$) and 0.6275 ± 0.0031 ($L = 8$), agreeing with the curve to within one standard deviation. No finite-shot point is shown at $L = 14$: its Born weights were not computed, and they are not extrapolated. **(c)** The absolute counts that panel (b) normalizes away, plus the third one. Sector dimension $\dim$ (grey) and the support $|\mathcal{S}|$ of the reported subspace (blue) both grow exponentially; the measured number of shots T that reaches the finite-shot operating point of panel (b) (green) is 2.4×10^4 , 2.6×10^5 , 3.1×10^6 , 1.9×10^7 at $L = 6, 8, 10, 12$ (s.d. over seeds 2.3×10^3 , 8.1×10^3 , 3.8×10^4 , 1.2×10^5). $L = 4$ carries no shot count: its 24-state sector is degenerate for this purpose and its budget is censored by the 10^8 -per-snapshot cap. Log-linear fits over the six sizes drawn in (c) (four for T , which has no $L = 4$ and no $L = 14$ point) give $\dim \sim e^{1.30L}$ ($R^2 = 0.9999$), $|\mathcal{S}| \sim e^{1.05L}$ ($R^2 = 0.998$) and $T \sim e^{1.12L}$ ($R^2 = 0.996$, four sizes). Those exponents are fits to the points this panel plots, not to the five-size fits stored in `honest_sampling.json` (1.108 and 1.082). Over $L = 6$ –12 the exponent of T exceeds that of the populated support by 0.054 per site ($[-0.066, 0.174]$) and that of the plotted support by 0.08 ($[-0.03, 0.19]$): these data do *not* establish that the shot budget grows with a larger exponent than the support, only that it is larger in absolute terms at every size measured (Table IV). **Resolution dependence.** Over $\eta/t \in [0.05, 0.50]$ the exponent of $|\mathcal{S}|$ moves from 1.165 to 1.017 and that of the fraction in panel (b) by a factor 2.2, from -0.1266 to -0.2743 (Table S5, $L = 4$ –12). In absolute value both move by the same 0.148: they are one exponent, and neither is stable.

TABLE V. The two collective scales, across four system sizes. Charge: the Mott gap $\Delta = \mu^+ - \mu^-$ from exact ($N \pm 1$) ground-state energies (half-filled Hubbard ring, $U/t = 8$; particle-hole symmetry $\mu^+ + \mu^- = U$ verified to 10^{-8} at every size, ARPACK checked against dense `eigh` on every sector of dimension ≤ 5000). $\Delta_\infty = 4.67952t$ is the Lieb–Wu thermodynamic value [53], evaluated here from its exact integral representation to an absolute quadrature error of 2×10^{-9} and cross-checked against a second exact representation, the two agreeing to 10^{-13} . Spin: the lowest pole of $S^{zz}(q = \pi, \omega)$, a pole position of the tridiagonal recursion and therefore independent of the broadening used to display it. Right-hand block: the same quantity in the pure Heisenberg ring at the effective exchange $J = 4t^2/U = 0.5t$, a model with no charge degree of freedom at all. The residual column is quoted to four decimals so that it equals the Δ/t column minus $\Delta_\infty/t = 4.67952$; at five it would not be.

L	charge		spin (Hubbard)		spin (Heisenberg, $J = 0.5t$)	
	Δ/t	$(\Delta - \Delta_\infty)/t$	Δ_s/t	$L \Delta_s/t$	Δ_{ST}/t	$L \Delta_{ST}/t$
6	5.35813	0.6786	0.34857	2.091	0.34237	2.054
8	5.18584	0.5063	0.23958	1.917	0.26134	2.091
10	5.02224	0.3427	0.19956	1.996	0.21162	2.116
12	4.96876	0.2892	0.16635	1.996	0.17792	2.135

Hubbard-band separation of Fig. 4, and every pole of $S(q, \omega)$ at $L = 12$ carrying at least 1% of the largest weight at its momentum—97.5% or more of the weight at each q —lies above it (Fig. 7); weaker poles are not resolved and are not bounded by this statement. No extrapolation exponent is fitted: a straight line in $1/L$ undershoots Lieb–Wu by $0.12t$, which is evidence that the form is not $1/L$, not evidence about the limit.

Spin. The lowest pole of $S^{zz}(q = \pi, \omega)$ gives $L\Delta_s = 2.091, 1.917, 1.996, 1.996$: a $1/L$ closing with a coefficient of about 2 (mean 2.000, range 1.917–2.091), and a fit $L\Delta_s = d_0 + d_1L$ has a slope consistent with zero ($d_1 = -0.010 \pm 0.018$). A linear-in- $1/L$ intercept of Δ_s is -0.020 ± 0.020 , against 4.561 ± 0.044 for the charge gap; but the pure Heisenberg ring, whose gap is known to vanish, gives $0.0142 \pm 0.0014t$ at these sizes, so a vanishing intercept is not by itself evidence that a gap closes, and the flatness of $L\Delta_s$ is what carries the statement. That Heisenberg ring at $J = 4t^2/U = 0.5t$ —a different Hamiltonian, Hilbert space and code path—reproduces the $1/L$ law, $L\Delta_{\text{ST}} = 2.054\text{--}2.135$, differing from the Hubbard value by 6.5% at $L = 12$ with a residual that changes sign: it confirms the scaling and the coefficient to within 10% and does not pin the coefficient (Sec. S7).

The two channels therefore behave *qualitatively oppositely* under a change of system size—one converging to a finite thermodynamic value, the other extrapolating to zero as $1/L$ —with a factor 29.9 between them at $L = 12$, and with $1/L$ intercepts that separate. That opposition, measured at four sizes with an independent control on the smaller scale, is the physical check this section makes. It is a statement about pole positions, not about lineshapes, so it does not depend on the broadening at which either response is plotted.

B. Resolution: why the statement rests on pole positions

A spin–charge statement cannot be read off a lineshape at a single size, for a reason of resolution. At $L = 12$ the spin scale is $0.16635t$ against the $\eta = 0.10t$ used for Fig. 8: the gap is 1.66η , i.e. 0.83 of the Lorentzian full width at half maximum. At that resolution a spin response with a $0.17t$ gap and one with none are not distinguishable by eye in a single-size plot. Terms that presuppose a resolved dense spectrum—“continuum”, “gapless” and “edge of the continuum”—would need that distinction, and none is used for any channel of this work at this size.

What establishes the physics instead is the four-size scaling of Table V, plotted in Fig. 6, and a pole census (Table S7): between $L = 6$ and $L = 12$ the sector grows by a factor 2440 while the largest per-momentum count of poles above 1% of the channel maximum grows from 3 to 7 in the addition branch, from 4 to 9 in the charge channel and from 2 to 3 in the spin channel, a discrete spectrum whose lines are broadened together rather than a dense one forming. The census, the sum rule as measured, and the size convention of the exact-diagonalization literature together with the dynamical-DMRG resolution criterion [80] this work does not meet at $L = 12$, are in Sec. S7.

C. Relation to the known spin–charge physics

That the spin and charge responses of the one-dimensional Hubbard model live on separate energy scales has been known exactly since Lieb and Wu [53], is textbook [81], was seen in angle-resolved photoemission thirty years ago [82, 83], is already legible in the spectral weight of twelve-site clusters embedded by cluster perturbation theory [84], and has been computed by dynamical DMRG since 2007 [3] in all three of the momentum-resolved channels reported here—in that reference’s notation the charge factor $N(q, \omega)$ at $L = 60$ (at $q \approx \pi$ up to $L = 120$), the spin factor $S(q, \omega)$ at $L = 100$ and the one-particle function $A(q, \omega)$ at $L = 90$, which are respectively our $S(q, \omega)$, our $S^{zz}(q, \omega)$ and our $A(k, \omega)$, for the extended Hubbard model of which ours is the $V = 0$ line; the same literature treats the Hubbard chain at $L = 48$ with $\eta = 0.1t$ by correction-vector DMRG, and computes the full Heisenberg-chain $S(k, \omega)$ at $L = 64$, with the $k = \pi$ cut to $L = 128$ [4]. Here it serves as a check, against a known answer, of the exact full-sector references against which the sampled reconstructions are scored, with the des Cloizeaux–Pearson lower boundary [54] and the Müller upper bound [85] recovered as thermodynamic-limit references rather than as findings. At $L = 12$ the spin peaks follow that lower boundary for $q \leq \pi/2$ and lie above it toward $q = \pi$ (Fig. 8).

IX. HARDWARE RUNS, CLASSICAL REACH, AND SCOPE

A. The two hardware runs, and what “exact by coverage” means

Two device runs were made, and they are the only quantum hardware used in this work. The first, on the IBM Heron processor `ibm_fez`, sampled the $L = 6$ periodic Hubbard ring at $U/t = 4$ with 12 qubits and 3.5×10^5 shots, and post-selection on particle number and S_z , with no configuration recovery, retained all 300 determinants of the $(N+1)$ sector, so the reconstructed A_{hw} equals the exact A *by coverage*: Rayleigh–Ritz on the whole sector is its diagonalization,

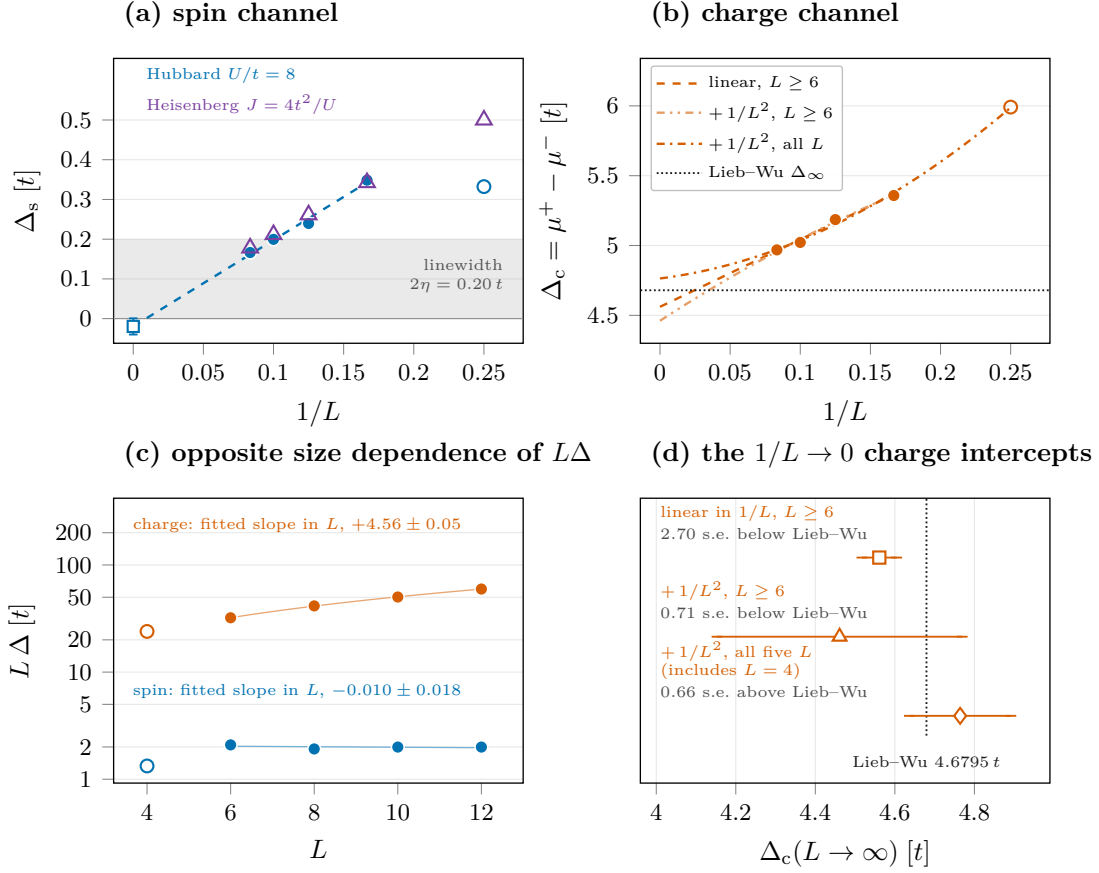


FIG. 6. The two gaps of the half-filled Hubbard ring scale in opposite directions. Periodic ring, $U/t = 8$, $L = 4, 6, 8, 10, 12$; in (a) and (b) the abscissa is $1/L$, so the sizes run $L = 12, 10, 8, 6, 4$ right to left, and open circles are $L = 4$, which the $L \geq 6$ fits exclude (the other open markers are not $L = 4$: the Heisenberg triangles of (a) are drawn open at every size, and the open square of (a) and the markers of (d) are fit intercepts). Every gap is an exact Lanczos pole, full reorthogonalization, *no* broadening. **(a)** Spin gap, lowest pole of $S^{zz}(q = \pi, \omega)$: $0.34857, 0.23958, 0.19956, 0.16635 t$ at $L = 6, 8, 10, 12$ ($0.33232 t$ at $L = 4$). Dashed: linear fit in $1/L$, $L \geq 6$ ($R^2 = 0.988$); the open square is its intercept $-0.0196 \pm 0.0204 t$, 0.96 s.e. from zero. Triangles: the pure Heisenberg ring $J = 4t^2/U = 0.50 t$, an independent control on a different code path, whose lowest $S^{zz}(q = \pi)$ pole reproduces its singlet-triplet gap to 8×10^{-15} . For $L \geq 6$ the two channels agree to within 8.3% (ratios 1.018, 0.917, 0.943, 0.935); at $L = 4$ they do not (0.665), which is why that size is excluded. Grey band: the linewidth actually used for the published spin figure, FWHM $= 2\eta = 0.20 t$ at $\eta = 0.10 t$. The $L = 12$ gap lies inside it ($0.16635 t = 0.83$ FWHM), and broadened by this η it still carries 26.5% of its peak height at $\omega = 0$. One size alone therefore cannot separate a closing gap from no gap. **(b)** Charge gap $\Delta_c = \mu^+ - \mu^-$ from the $N \pm 1$ sectors: $5.3581, 5.1858, 5.0222, 4.9688 t$ at $L = 6, 8, 10, 12$ ($5.9914 t$ at $L = 4$). At $L = 12$ that is $27.6\eta = 13.8$ FWHM for the $\eta = 0.18 t$ of the one-particle channel, 16.6 times further from $\omega = 0$ in units of its own linewidth than the spin gap. Three extrapolations are drawn and they disagree. The ordinate omits zero because the question here is which finite value the gap tends to, not whether it vanishes. **(c)** The same data with no extrapolation model, logarithmic ordinate: $L\Delta_s$ is flat over $L \geq 6$ (mean $2.0 t$, the four values spanning 8% and not monotone, so no third figure is quoted; fitted slope in L -0.010 ± 0.018 , 0.57 s.e. from zero) while $L\Delta_c$ grows linearly ($+4.56 \pm 0.05$, 101 s.e.). The thin lines are those two linear fits, curved here only because the ordinate is logarithmic. **(d)** Those three intercepts on a scale where their standard errors are legible; on the scale of (b) the $\pm 0.0439 t$ bar is thinner than its marker. Linear, $L \geq 6$: $4.5609 \pm 0.0439 t$, undershooting the exact Lieb-Wu gap $4.6795 t$ by $0.1186 t$, 2.70 s.e. Same protocol plus a $1/L^2$ term: $4.4611 \pm 0.3084 t$, undershooting by $0.2184 t$ (0.71 s.e., standard error 2.4 times larger). Only the $1/L^2$ fit over all five sizes brackets Lieb-Wu ($4.7641 \pm 0.1278 t$, 0.66 s.e. above), and it is the one fit here that includes $L = 4$, the size (a) and (b) mark anomalous. These sizes therefore fix the $L \rightarrow \infty$ charge gap only to about ± 0.1 – $0.3 t$: compatible with Lieb-Wu once a $1/L^2$ term is allowed, not evidence for it. What all three exclude is zero, the weakest by 14.5 s.e. and the linear one by 104 s.e. *Caveat, because it limits (a).* A vanishing intercept is not by itself the evidence: the Heisenberg control, whose gap is known to vanish in the thermodynamic limit, has a linear-in- $1/L$ intercept $0.0142 \pm 0.0014 t$, 10.0 s.e. above zero, and its own $L\Delta$ drifts $+6.8\%$ from $L = 4$ to $L = 12$, both from its logarithmic corrections. The model-free statement is (c).

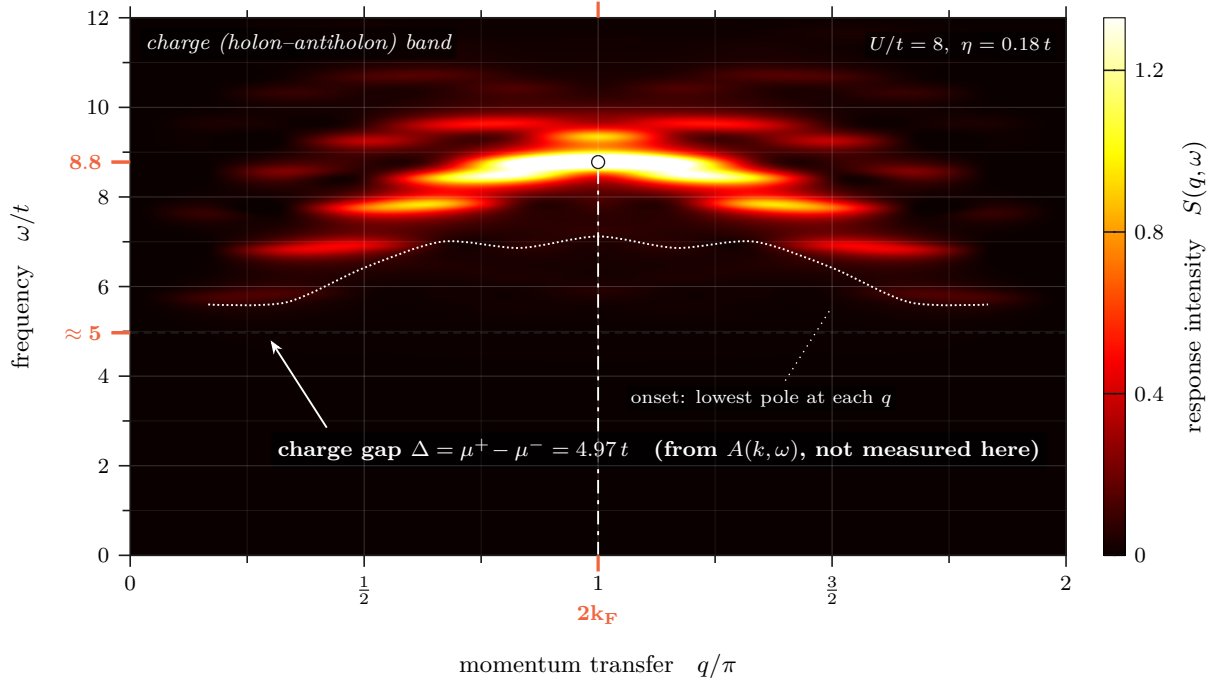


FIG. 7. **Dynamical structure factor: the density response.** $S(q, \omega)$ of the half-filled periodic Hubbard ring ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18 t$)—the density–density response, a number-conserving (neutral) excitation, computed by a Haydock continued fraction of depth $n_\ell = 200$ in the N -particle sector—a finite-depth quadrature of the spectral measure, not the Lehmann sum (Sec. S7). Color is the *response intensity* $S(q, \omega)$ (units $1/t$): bright means the system responds strongly at that (q, ω) , dark means it does not. The weight forms the two-particle charge (holon–antiholon) band, whose dotted curve is the onset, the lowest pole resolved at each q , and sets the low-frequency scale of the band and traces its lower-boundary dispersion; the diffuse upper boundary is left unmarked. At $L = 12$ this band is six to nine discrete poles per momentum merged by the broadening, not a dense spectrum (Table S7). Two features are decisive. (i) The band is gapped on the scale of the charge gap: every resolved pole lies above $\Delta = \mu^+ - \mu^- = 4.97 t$ (exact $(N \pm 1)$ ground-state energies), the lowest onset reached here being $5.60 t$ at $q/\pi = 1/6$. The limit $q \rightarrow 0$ is inaccessible to this observable, $S(q=0, \omega>0)$ vanishing identically by density conservation, so Δ is drawn as a horizontal reference level imported from the single-particle channel and nothing is claimed about that limit. It is the *same* Mott gap that separates the Hubbard bands in the single-particle channel (Fig. 4): two independent dynamical observables, one correlation gap. (ii) The band peaks at $q = 2k_F = \pi$ at $\omega = 8.78 t$ (coral tick, at the measured peak), within 10% of the bare Hubbard scale $U = 8 t$ of a doubly-occupied site; the tick marks the peak, not U . The reflection $S(q) = S(2\pi - q)$ holds to $\sim 10^{-4}$ (Lanczos precision), and the analytic f-sum rule $\int \omega S(q, \omega) d\omega \propto (1 - \cos q)$ is recovered by the plotted (finite-window) grid to $\sim 0.3\%$. This is the collective observable the same bitstring-sampled primitive targets; as for $A(k, \omega)$, faithfully sampling the *full* response carries the sector-fraction cost of Sec. VII. This channel carries no shot budget: it is a full-sector reference (Table I).

and the vanishing error is an identity rather than a measurement of device fidelity (Fig. 9(a)). The circuits prepared a computational-basis $(N+1)$ determinant rather than $\hat{c}_p^\dagger |\Psi_0\rangle$ and applied $R_{xx}R_{yy}$ hopping without Jordan–Wigner strings, so they are not the primitive of Fig. 1. The post-selection, moreover, read each bitstring in reversed qubit order, which reflects the ring and exchanges the two spin species: under that reading none of the 3.5×10^5 shots of a noiseless simulation of the same seven circuits is retained, against all of them under the correct reading, so every one of the 300 retained determinants arose from a device error (`src/hw_bitorder_check.py`). The run is therefore an execution record: the steps of the pipeline execute end to end on a processor, but the run does not show that the device sampled the intended distribution, and it establishes nothing about accuracy at complete or incomplete coverage. The second, stretched N_2 at 24 qubits on `ibm_marrakesh`, reaches 0.59 ± 0.14 mHa of the exact active-space energy after recovery, below the 29.5 mHa of the noiseless simulation of the same circuit (Fig. 9(b)). The record does not store the recovered subspace sizes on which that comparison turns, so it cannot say whether device noise helps, although a study under a simple noise model reports the mechanism [86]. That study also finds that recovery from randomly generated configurations alone builds accurate CI spaces, and no such control was run here. What is controlled is recovery against no recovery on the same corrupted bitstrings, and there the gain grows with the noise (Figs. S7 and S8). Neither run is a spectral record in the sense of Refs. [11, 12, 15], whose spectra are determined by device data; the full account, the

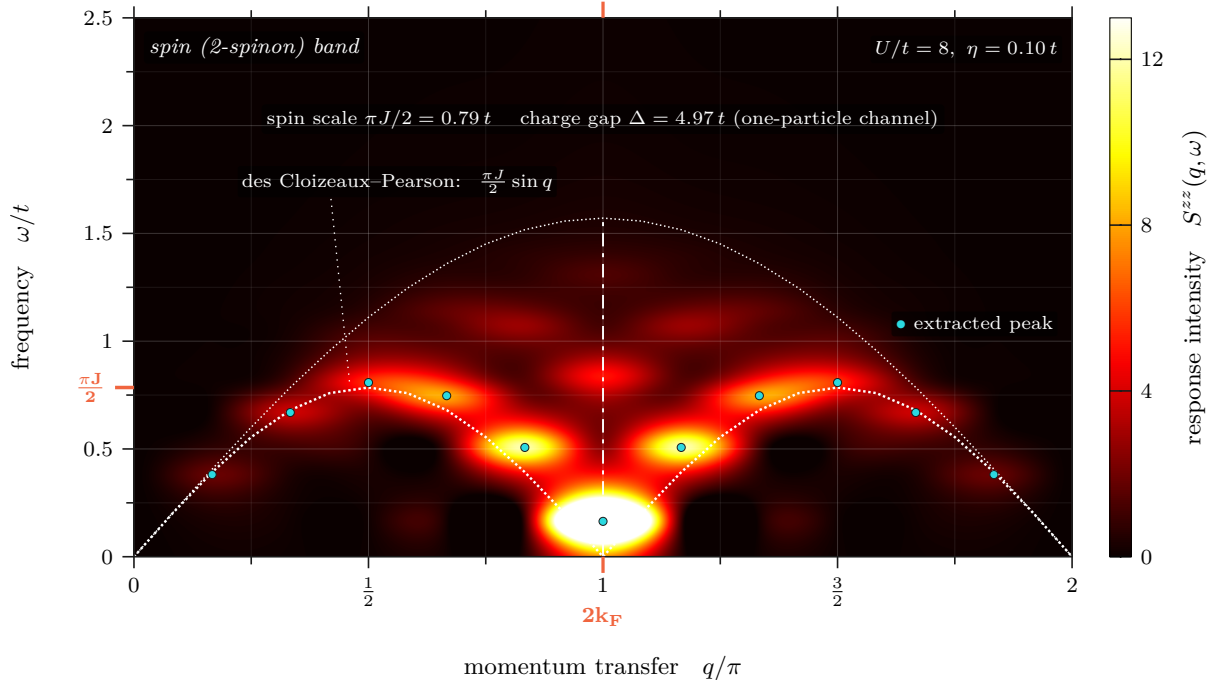


FIG. 8. **Spin structure factor: the low-frequency companion of the charge response.** The dynamical spin structure factor $S^{zz}(q, \omega)$ of the same half-filled periodic Hubbard ring ($U/t = 8$, $L = 12$; here $\eta = 0.10 t$, finer because the spin scale is small), computed by a Haydock continued fraction of depth $n_\ell = 200$ from a spin seed $S_q^z|\psi_0\rangle$ in the N -particle sector; at this η a reference of that kind is not converged in general (the single-particle one moves by 3.0% between $n_\ell = 100$ and 200, Sec. S6), so the lineshape is quoted with its depth and not as exact (Sec. S7). *Contrast Fig. 7:* the charge scale stays at the Mott gap $\Delta \approx 5 t$ at every size measured, while the spin scale is $0.16635 t$ at $L = 12$ and closes as $\approx 2/L$ (Table V). At this size the spin response is one dominant pole plus at most two satellites per momentum (Table S7), not a dense band, and what the figure shows is a set of broadened lines; whether the band touches $\omega = 0$ in the thermodynamic limit is not established here and is not claimed. This is spin-charge separation in the collective (two-particle) channel: charge and spin live in energetically separate worlds, the spin set by the superexchange $J = 4t^2/U = 0.5 t$, some $16\times$ smaller than U . The spectral weight piles up on the des Cloizeaux-Pearson [54] lower boundary $\frac{\pi J}{2}|\sin q|$ (dotted), bounded above by $\pi J|\sin \frac{q}{2}|$ [85]; the numerically-extracted peak dispersion (cyan markers: the maximum of each broadened line on the $0.006 t$ frequency grid) tracks this lower boundary to within about 3% for $q \leq \pi/2$ (and $q \geq 3\pi/2$); toward $q = \pi$ it lies above it, by 10% at $q = 2\pi/3$ and 29% at $q = 5\pi/6$, and at $q = \pi$, where the boundary vanishes, it sits at the finite-size spin gap; the antiferromagnetic soft mode at $q = \pi$ carries the dominant weight. The reflection $S^{zz}(q) = S^{zz}(2\pi - q)$ holds to double-precision round-off and $S^{zz}(q \rightarrow 0) \rightarrow 0$ (total- S^z conservation). The same bitstring-sampled primitive that targets $A(k, \omega)$ and $S(q, \omega)$ targets this collective response as well. This channel carries no shot budget: it is a full-sector reference (Table I).

job identifiers and the shot arithmetic are in Sec. S8.

The molecular suite. On the seven Hamiltonians of the nineteen-molecule suite (Sec. S15.1, Table S11) whose sectors have dimension 20 to 300, the same primitive reproduces the orbital-projected $A(\omega)$ to a relative L_1 between 2.2×10^{-6} and 1.0×10^{-4} from 21% to 50% of the sector, a consistency check of the primitive outside the Hubbard family and not a scaling statement. The other twelve reconstructions are exact by construction, because their sampled subspaces, of at most twelve determinants, already contain the probe's Krylov space, so the suite cannot show that the reconstruction is efficient across chemistry.

B. What the cost tracker implies for classical simulation

The cost tracker of Sec. VI, the bond dimension χ , is by definition the cost parameter of a matrix-product representation of the same state: wherever it says the determinant support is small, a matrix-product state is small too. A sampler can therefore be needed only for a target whose minimal bond dimension is large in every ordering while the sampled support stays compact, a property of geometry and orbital connectivity that two-dimensional lattices and molecular orbital graphs would test and that nothing here measures (Sec. S8). Every target measured

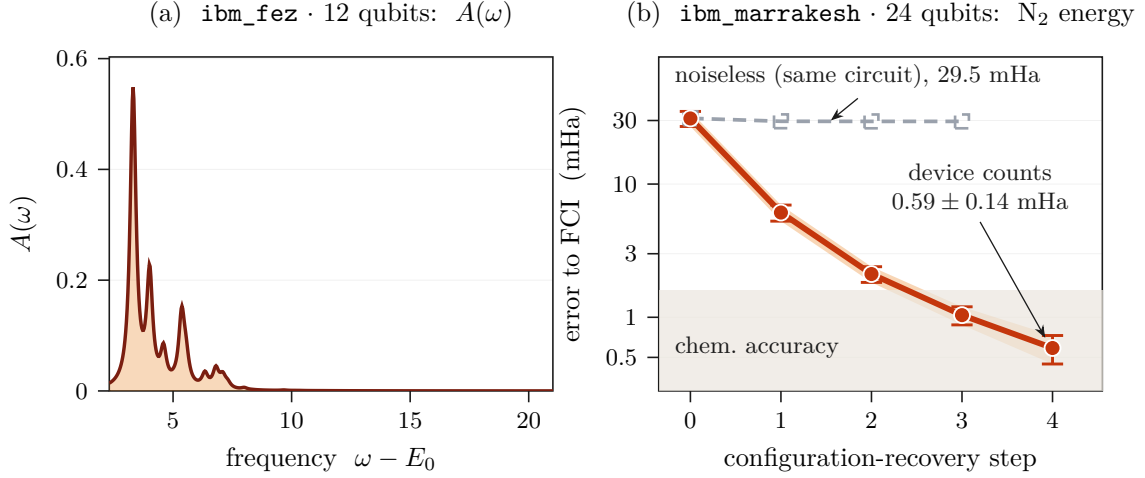


FIG. 9. **The complete quantum-hardware record of this work: two runs on IBM Heron processors.** (a) Electron-addition branch of the local spectral function, $A_{0+}^+(\omega)$, of the $L = 6$ periodic Hubbard ring at $U/t = 4$, $\eta = 0.15t$ (12 qubits; the classical validation of Fig. 2 is a different instance, $U/t = 8$), computed on the subspace post-selected from 350,000 computational-basis bitstrings—50,000 from each of seven circuits—measured on **ibm_fez** (measurement and Pauli twirling and dynamical decoupling; no readout-error mitigation and no configuration recovery). The retained shots covered the complete $(N+1)$ sector ($|\mathcal{S}| = 300/300$), so the reconstruction is exact by coverage; they were post-selected in reversed bit order, so every retained shot is a device error (Sec. IX A). The panel is an execution record, and neither a fidelity measurement nor a claim of advantage. (b) Ground-state energy error of stretched N_2 ($R = 2.0 \text{ \AA}$, CAS(10e,12o)/cc-pVDZ, 24 qubits) versus self-consistent configuration-recovery step, on **ibm_marrakesh** (10,000 shots, CCSD-seeded local unitary cluster-Jastrow (LUCJ) circuit). Recovery on the device counts (solid) reaches 0.59 ± 0.14 mHa of the exact FCI value—well below chemical accuracy—whereas the noiseless statevector simulation of the *same* circuit (dashed) is stranded at 29.5 mHa (its recorded recovery history ends at step 3, unchanged since step 2). That contrast is not read here as device noise *helping*: the mechanism such a reading needs is decided by the two recovered subspace sizes, and the record stores neither (Sec. IX A). The solid curve is the mean over 8 recovery seeds on the cached device counts; the shaded envelope and the capped whiskers at every step both show the ± 1 population s.d. (normalized by 8, not 7) across those 8 seeds (widest at step 0, 31.1 ± 4.0 mHa, where it starts together with the noiseless curve, and tapering as recovery converges to 0.59 ± 0.14 mHa; best seed 0.40 mHa). No random-bitstring control was run, so the panel cannot separate what the device counts contributed from what recovery alone supplies; the controlled recovery-versus-no-recovery comparison, under a simulated bit-flip model, is Fig. S8. These two runs are the entirety of the quantum hardware used here; every other result in this work is an exact classical simulation.

here—Hubbard rings at $L \leq 14$, nineteen active spaces, and a largest minimal bond dimension of 42 over the 74 deposited exact ground states—lies on the other side: there is no computation performed here that a classical method cannot perform, and we make no advantage claim at any size reached. The literature shows the same pattern. The 7260 trajectories of a 120-qubit Fermi–Hubbard dynamics experiment [87] were reproduced by tensor contraction at bond dimension 32 [88], the pattern in which tensor-network and Pauli-propagation methods have overtaken earlier hardware demonstrations [89–91], with Pauli propagation since extended to real-time correlators [92]; the energy of any single-layer unitary cluster-Jastrow circuit, the largest (77-qubit) experiment of Ref. [35] included, is now computed classically in polynomial time [93], and an extended matchgate simulator evaluates, to additive approximation, the Born-rule probabilities of hardware-sampled cluster-Jastrow bitstrings at a cost exponential only in the magnitudes of the controlled-phase angles [94]; on the ground-state energy the strong classical selectors already win on cost [95, 96], and even optimally ordered SQD subspaces need a number of configurations that grows exponentially with lattice size for Heisenberg and Hubbard ground states [97].

A dequantization test. On an emulated, under-sampled head for N_2 ($R = 2.5 \text{ \AA}$; 1000 shots drawn from the exact FCI marginal and corrupted by a synthetic Heron-class noise model, ten paired seeds; no device data), a classical Epstein–Nesbet prior [98] applied to the same histogram cuts the energy error by 36%, from 38.22 to 24.38 mHa (paired $t_9 = 12.37$, all ten seeds), while a generative flow network [99] fused onto that prior moves it a further 1.80 mHa, a difference ten paired seeds do not resolve ($t_9 = 1.19$) and whose sign reverses between two five-seed halves; at 500 shots the fused generator is worse than the classical control (Sec. S16). On this instance the dominant determinants are supplied by the sampled histogram (emulated, so the test does not measure whether a device state would supply them) and completed by a classical prior rather than manufactured by a learned generator, and the largest improvement available to an under-sampled head costs no quantum resources at all.

C. Scope

What the work establishes is four things, each with its measurement attached: one sampling primitive that targets four frequency-resolved channels, of which the single-particle ones are reconstructed here from ranked or sampled subspaces and the two neutral ones are full-sector references (Table I); a two-sided bound on the error of the reconstruction, whose leakage term is a posteriori, with its non-vacuity threshold measured at five sizes up to a sector of dimension 10 306 296 and its crossing calibrated to a true relative error of $\approx 3 \times 10^{-3}$ within a factor of three (Sec. V); an obstruction showing that no orbital-rotation invariant supplies an a priori predictor of that bound useful on every state (Sec. VI); and three prices—support, resolution and shots—each measured here (Sec. VII).

The momentum-resolved spectra are one-dimensional Hubbard rings at $L \leq 12$ with $\eta = 0.10\text{--}0.18t$ (the support, certificate and shot measurements reach $L = 14$), which is the exact-diagonalization convention for this observable—Xu, Lu, Tohyama and Shao compute it at $L = 12$ and $\eta = 0.2t$ at half filling, and at $L = 16$ at quarter filling, in the extended Peierls–Hubbard chain [100]—but the resolution criterion the dynamical-DMRG literature applies at those sizes is not met here, so Sec. VIII rests its spin–charge statement on pole positions at four sizes rather than enlarge η .

D. Data availability statement

The data that underlie the figures and the quoted numbers of this work are openly available in the reproduction repository [101] at <https://github.com/nicolasbonilla/dynamical-spectral-functions-sqd>, archived on Zenodo under DOI 10.5281/zenodo.22967220, the code under the MIT license and the manuscript, the deposited datasets and the reproduction documentation under CC BY 4.0. Every figure’s plotted values are deposited—as a JSON table in `data/` or as a plain-text table in `paper/figs/`—and `docs/FIGURE_PROVENANCE.md` names the file that is authoritative for each; the one exception is Fig. 1, a circuit schematic that plots no data. The classical data—the exact-diagonalization spectral functions of the one-dimensional Hubbard model, the molecular suite and the resource statistics—are additionally *regenerable*: the computation scripts that produced them are deposited in `src/`, and `docs/REPRODUCE.md` gives the reproduction commands together with a measured table of what a clean clone does and does not reach. An automated guardian (`python src/verify.py`) recomputes the underlying physics by exact diagonalization and checks the deposited artifacts against it—1280 checks over 14 563 numeric assertions, exiting non-zero on any discrepancy. It covers the deposited data files and the numbers printed in the figure sources, in the tables and in the abstract; it is not a proof that every sentence of this manuscript is verified, and four items are printed by name on every run, the known open defects; the ratios plotted in Fig. S5 are re-derived by it from the deposited certificate rows.

The quantum-hardware results are deposited as the arrays that enter the figures, together with the backend name, the shot count and the IBM Quantum job identifier for each of the two runs (`data/heron_spectral.json`, `data/hw_lucj_n2_result.json`). *The raw measured bitstrings are not part of the deposit*: the counts can be obtained only from the IBM Quantum service, by retrieving the recorded job identifier (`service.job(...)`) in place of the submission cell of the notebooks in `notebooks/`, which requires an account and a job whose results are still retained, so the classical post-processing of the raw counts cannot be re-run from the deposited files alone. The hardware runs themselves cannot be regenerated in any case—the jobs are closed, and a device executed today has a different calibration, so new counts would be new data rather than a reproduction. The dequantization control of Sec. S16 is regenerated seed by seed by `src/gflow_dequant.py`, which needs `pyscf` and `torch` and writes `data/gflow.json`. The $L = 14$ ground-state checkpoint (94 MB), on which the largest vacuity factor and the $L = 14$ support count rest, is deposited in `ckpt/` with its SHA-256 checksums. Known gaps between the deposited code and the deposited artifacts are enumerated in `docs/KNOWN_DISCREPANCIES.md`; those still open are the rasters embedded in Figs. 4, 7 and 8, which are deposited as rendered and cannot be regenerated from the deposit; the third panel of Fig. 5 and the captions of Figs. 3 and 6, edited by hand after generation; two computation scripts without deposited output; and the parts of the pipeline that require `pyscf` or an IBM Quantum account.

X. CONCLUSION

For a spectral function reconstructed by Rayleigh–Ritz on a sampled subspace, the weight the subspace missed, together with the Hamiltonian coupling across its boundary, bounds the error. The captured Born weight alone does not: an analytic counterexample and a counting argument rule out any bound indexed on it (Sec. IIID). The weight a subspace keeps is displaced in ω , not discarded. On the probe’s own Born support, where the captured weight is exactly one, the relative error at $\eta = 0.18t$ is therefore still 0.43 at $L = 6$ and 0.35 at $L = 8$. Theorem 1 turns the answer into an a posteriori certificate for any orthogonal projector, given the exact ground state. It bounds the L_1 error from below

by the Born weight the subspace failed to capture, and from above by the smaller of the trivial bound and a leakage branch, built from that weight and the coupling that leaks across the subspace boundary at the resolution asked for. That coupling is computed from the Ritz pairs and the retained part of the probe, without the exact spectral function. What a Krylov-adequate subspace does guarantee in addition is the classical moment exactness of Krylov projections, through order $2K + 1$ and no further, a reach stated exactly and measured on coordinate subspaces (Theorem 2).

Evaluated on the scanned subspaces at five system sizes, up to a sector of dimension 10 306 296, the certificate is vacuous at every operating fraction of the resource scan. There its leakage branch exceeds the trivial bound by factors of 1.66 to 8.61, a verdict converged in Krylov depth at $L = 12$. The certificate becomes informative only well above those fractions, and there it is calibrated. Over sixteen (L, η) cells at $L = 6$ –12, it first improves on the trivial bound at a true relative L_1 error between 1.2×10^{-3} and 7.3×10^{-3} . That spread is a drift, not scatter: it is ordered in both L and η and conservative in L . The certificate locates the crossing without the exact spectral function, given the exact probe. A two-constant rule in the missed weight $1 - w_S$, fitted on some sizes and tested on the others, localizes the same crossing more tightly on eleven of the fourteen splits. What distinguishes the certificate from such a rule is that it is a proven inequality whose verdict still holds when the captured weight w_S is replaced by an assumed lower bound on it. The Born-ranked, infinite-shot reconstruction of Fig. 3, at 85% of its sector, is non-vacuous in all sixteen of its momentum channels. Its relative bound there, 0.77–1.75 against the trivial 2, is 350–900 times its true error.

The leakage depends on the basis, and that dependence is measured. In the determinant basis in which a device measures, the coordinate support of the Krylov space $\mathcal{K}(H, \phi)$ is every determinant the reflection about the seed site allows, measured at $L = 6$ to 14. At full captured weight, a subspace holding the whole probe therefore leaks unless it contains all of them. The vacuity at the operating fractions, all below full captured weight, is measured and not deduced from that fact. A momentum-resolved probe measured in plane-wave orbitals has its Krylov space inside one momentum block instead, $1/L$ of the sector. On that block the certificate still needs 455–483 of 490 determinants at $L = 8$, and the block, like the sector, grows exponentially with L (Sec. VE). No orbital-rotation invariant bounds the boundary leakage *usefully* on every state in advance: an invariant of the occupation spectrum cannot see a boundary. On pairing-model fibers every such bound is vacuous, while the subspace-aware certificate still separates a non-vacuous value from the trivial bound (Proposition 2). The one-body non-Gaussianity $\mathcal{F}_1$ is constant along a geminal orbit that contains a point of classical cost $O(K)$ for K geminals, at bond dimension 2 and 2^K determinants in the pairing basis (Theorem 3). The orbit also contains points at which sampling is plausibly hard, a hardness not established here. This is consistent with the separation drawn by Tarabunga *et al.* [17] between covariance-matrix quantities, two of which lower-bound the non-Gaussian gate count, and full-state natural-orbital participation entropies, which upper-bound the classical simulation cost in an orthonormal Gaussian basis.

The certificate does not certify the reconstructions at the operating fractions of the resource scan, and the regime in which it would certify them costs more shots than the method saves. The method is priced in support, resolution and shots (Sec. VII). The determinant support that reconstructs $A(\omega)$ to relative $L_1 < 0.05$ at $\eta = 0.15t$ grows as $e^{1.05L}$ over $L = 4$ –14, and the shot budget grows from 7.8×10^2 to 1.9×10^7 over $L = 4$ –12. Certification costs more: moving from the operating fraction to the non-vacuous band multiplies the shots per snapshot needed for the marginal configuration by 7.5 to 10 at $L = 8$ and by 20 to 35 at $L = 12$. At the acceptance criterion this work uses elsewhere, relative $L_1 < 0.05$, the certificate needs at least 0.98 of the sector at $L = 8$, a fraction at which one would diagonalize the whole sector instead. On the scanned grid of fractions it does not reach that criterion at $L = 6, 10$ or 12.

In the bound, the frequencies of both spectra are measured from the exact ground-state energy E_0 (hypothesis H1 of Theorem 1). In a shot-based pipeline E_0 is itself estimated on a sampled subspace, and the bound then holds only up to an additive term the certificate does not control. For the energy shift of $6.7 \times 10^{-4}t$ obtained from 60% of the N -electron sector at $L = 8$, that term moves the bound by +0.20%, while the measured error grows by a factor of 16 (Sec. S11). The probe is likewise built from the exact ground state (hypothesis H4). With an approximate ground state instead, the bound applies to that probe’s own spectral function, whose distance to the exact one nothing computed here controls; no certificate in this work has been evaluated with an approximate probe. The full-sector references reproduce the opposite finite-size scaling of the two gaps: the charge gap decreases to within 6.2% of its Lieb–Wu value at $L = 12$, while the spin gap closes as $\approx 2/L$, with an independent effective-Heisenberg control (Table V). All spectra here are at exact-diagonalization sizes, the largest momentum-resolved one at $L = 12$. By comparison, dynamical DMRG computed the same three channels at 60 to 120 sites in 2007 [3], and correction-vector DMRG reaches $L = 48$ in the Hubbard chain [4]. In one dimension tensor networks are near-optimal [102, 103], and the largest minimal bond dimension over the 74 exact ground states deposited here is 42. No quantum advantage is claimed at any size reached (Sec. IX).

Whether a certified regime exists at low sector fraction can now be settled by a well-posed measurement, and the calibrated crossing is the instrument for making it. The answer turns on two open questions. At full captured weight, the identity behind Theorem 1 yields a certificate of second order in the boundary coupling. The first question is whether it has a counterpart below full weight, where the operating fractions lie, that can be evaluated without the exact spectral function. The second is whether some single-particle basis gives $\mathcal{K}(H, \phi)$ a coordinate support that is a

small fraction of the sector. The plane-wave orbitals are one such basis for a momentum probe, at about $L(L-1)$ further two-qubit gates whose effect on the sampled distribution is unmeasured; whether the support fills the block at $L \geq 10$ is unmeasured as well. Toward both, this work provides a two-sided bound whose boundary-leakage term is computable from the Ritz pairs and the retained probe, and a measured map of where that bound is informative and at what cost. It also provides the coordinate support that fixes, basis by basis, what zero leakage at full captured weight requires. For a sampled pipeline validated at exact-diagonalization scale, where the exact ground state is in hand, the captured weight is therefore not a convergence diagnostic for the spectral function built on the subspace. The error of that spectral function is bounded, though not estimated, by the weight the subspace missed together with the boundary coupling. The coupling is computable there from the same Ritz pairs, the retained probe and one matrix-vector product per Ritz vector, and the missed weight from the norm of the exact probe and its retained part.

ACKNOWLEDGMENTS

We acknowledge the use of IBM Quantum services for this work; the two device runs reported in Sec. IX A were executed on `ibm_fez` and `ibm_marrakesh`. The views expressed are those of the author and do not reflect the official policy or position of IBM or the IBM Quantum team.

AI tools were used in the preparation of this manuscript; the author takes full responsibility for its content.

Appendix A: Proof of the leakage certificate and of its closed form

This appendix proves Theorem 1 and the second equality of Eq. (6).

Proof. Write $u = R(z)\phi$ and $u_P = R_P(z)\phi_P$, extended by zero outside $\text{ran } P$.

(1) *The split.* Since $u_P \in \text{ran } P$ and $(z - H_P)u_P = \phi_P$ there, $(z - H)u_P = zu_P - PHu_P - QHu_P = \phi_P - QHu_P$; subtracting this from $(z - H)u = \phi$ gives

$$u - u_P = R(z)[\phi_Q + QH R_P(z)\phi_P]. \quad (\text{A1})$$

This is the Feshbach–Löwdin partition of the resolvent [55, 56], written as an exact identity rather than as an expansion: the first source term is the part of the probe the subspace never held, the second is the part of the Hamiltonian that carries amplitude out of it. Only $P^2 = P$, $P^\dagger = P$, $H = H^\dagger$ and $\eta > 0$ are used; no property of the determinant basis, of the ranking, or of the sampler enters anywhere below.

(2) *Pointwise.* $A - A_P = (\eta/\pi)(\|u\| + \|u_P\|)(\|u\| - \|u_P\|)$ and $|\|u\| - \|u_P\|| \leq \|u - u_P\|$, so $|A - A_P| \leq (\eta/\pi)(\|u\| + \|u_P\|)\|u - u_P\|$ at every ω .

(3) *Integration.* By Cauchy–Schwarz in ω ,

$$\|A - A_P\|_1 \leq \frac{\eta}{\pi} \left[\int (\|u\| + \|u_P\|)^2 d\omega \right]^{1/2} \left[\int \|u - u_P\|^2 d\omega \right]^{1/2}.$$

The first bracket is at most $(\pi/\eta)^{1/2}(\|\phi\| + \|\phi_P\|)$, by Minkowski in $L^2(d\omega)$ and the two sum rules. For the second, apply Minkowski to Eq. (A1): its first term obeys the sum rule with seed ϕ_Q , contributing exactly $(\pi/\eta)^{1/2}\|\phi_Q\|$; its second obeys $\|R(z)v\| \leq \eta^{-1}\|v\|$, since H is self-adjoint and $\text{Im } z = \eta$, and the definition (6) then contributes at most $(\pi/\eta)^{1/2}\Lambda_P(\eta)/\eta$. Multiplying the three factors gives the leakage branch of (7).

(4) *The other two branches.* A and A_P are nonnegative with known integrals, so $\|A - A_P\|_1 \leq \int A + \int A_P = \|\phi\|^2(1 + w_P)$ under no hypothesis at all; and $|\int A - \int A_P| \leq \|A - A_P\|_1$ gives the lower bound $\|\phi\|^2(1 - w_P)$.

(5) *The closed form.* $R_P(z)\phi_P = \sum_m c_m(z - \tilde{E}_m)^{-1}|\tilde{m}\rangle$, so

$$\|QHR_P(z)\phi_P\|^2 = \sum_{m,m'} \frac{c_m \bar{c}_{m'} \langle g_{m'}, g_m \rangle}{(z - \tilde{E}_m)(\bar{z} - \tilde{E}_{m'})}.$$

With $x = \omega + E_0$ the remaining integral is $\int dx [(x - \tilde{E}_m + i\eta)(x - \tilde{E}_{m'} - i\eta)]^{-1}$, which has one pole in each half plane; closing above picks up $2\pi i/[(\tilde{E}_{m'} - \tilde{E}_m) + 2i\eta]$, and multiplying by η/π gives Eq. (6). $\square$

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